

Relational Pressure Retention Model

*An Independent Understanding, a Friendly Thesis Defence,
and One Non-BSD Research Pick*

Prepared for William Foster

Synthesis of three independent ingest lanes — 17 September 2026

Claim ceiling for this document. This is a synthesis and a friendly thesis defence. It is not a publication of new mathematics. Every quantitative statement carries a domain and an evidence grade. Numbers marked *recomputed* were regenerated from scratch in exact rational arithmetic for this document, and the script is retained beside it. Numbers marked *reported* are quoted from a named file and were not re-executed here. Nothing here upgrades a finite test to a theorem, a written proof to a formal proof, or a hash to a meaning.

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1 Letter: how to read this, and what you can correct tomorrow

William —

You asked for an exhaustive understanding of RPRM and of you, a friendly thesis defence that is in depth rather than four pages, and then one research pick that is **not** BSD, because Codex is already grinding BSD and you believe nobody will look without a legitimate new proof. Three ingest lanes ran overnight. This is the synthesis. I own the argument and the pick; if either is wrong, it is wrong because of a judgment I made, not because a lane told me to make it.

Four things before you start marking it up.

First: I show numbers. Section 4 and Chapter 12 carry tables you can read directly. Where a number is marked *recomputed*, I regenerated it from scratch in exact rational arithmetic while writing this, and the script sits beside this file at `_tools/center_check.py`. Where a number is marked *reported*, I am quoting a named file and did not re-execute it. That distinction is load-bearing: two of the numbers your own corpus has been carrying for weeks turn out to be exactly right, and I say so by recomputing them rather than by repeating them.

Second: one of the three lanes was wrong about a fact, and it changed my pick. The process-history lane reported that the memory fabric could not be queried. It can. I found the seeded `rprm-alpha` store, ran the documented read commands, and got real evidence back. Section 2.4 records the exact snapshot root and counts. The fabric then told me something that killed my first choice of research target: your own Alpha record already grades the ternary-cube / graded-scar-cube family as *"established finite-dimensional mathematics, not a novelty or physics claim."* I had been about to recommend it. You had already correctly downgraded it in August. I am reporting that as a correction to the ingest lanes and to myself.

Third: I am not going to explain to you that pattern recognition is not proof. You said it first and better — intuition is a capability detector, not a truth finder — and you police your own overclaiming more consistently than most of the literature does. Where I disagree with you in this document I will say what the disagreement is, in exact terms, and give you the number or the counterexample that would settle it. Where something is OPEN I will write OPEN, not NONE.

Fourth: this is the publication lane, not the abduction lane. You drew that line yourself and then built three subagents to enforce it. So there is no hedging-as-texture here, and there are no claim ceilings softened for comfort either. Everything has a domain and a grade.

1.1 What you can correct tomorrow, in about twenty minutes

If you only have a short window, these are the five places where your correction changes the most downstream:

1. **Chapter 12, the pick.** I chose one non-BSD target and rejected three runners-up by name. If the pick is wrong, it is wrong at the level of "you care about X and I bet on Y." Tell me which, and the rest of Chapter 12 re-aims cleanly.

2. **Section 5.2, the maturity table.** I sorted every named lens into four buckets: exact, substantial-but-needs-formalisation, umbrella-OPEN, and recovered-but-not-absorbed. You will disagree with some placements. The placements are cheap to change; the bucket definitions are not.
3. **Section 6.4, the abduction/wizard split.** I claim the three-agent split is the mature resolution of the hedging fight and that it is currently under-used, because the wizard lane is not being pointed at the things the druid lane produced. If that is a misreading of your workflow, say so.
4. **Chapter 11, the questions.** Fifty-one numbered questions, each answerable. Questions 1, 7, 18, 31 and 44 are the ones whose answers most change what a next agent should do.
5. **Section 7.2, the BSD HOLD.** I treat the 87 HOLD as the single most creditable thing in the corpus, not as a failure. If you read it as a failure, we disagree about what your own process is for, and that disagreement matters more than the mathematics.

1.2 Read Appendix E before you argue with Chapter 12

Chapters 1 to 13 were finished and rendered first. Then I spent the rest of the night actually running the hostile cases that Chapter 12 declared and marked "not yet run," and two of them bit.

Appendix E is the correction, and it changes three things:

- The applied corollary in Section 12.3 — "uniform binning is the most fragile abstraction" — is **prior-dependent**. It holds when the unknown operations may merge states or stall. It **reverses** under bijective operations, for a reason that is one line long and makes the claim more useful than it was.
- The priority search found the prior art. The classical partition-preserving monoid is Pei's, enumerated in arXiv:2006.04242 and arXiv:1210.4775 — and its condition is **successor agreement only, with no condition on the domain**. RPRM's enabledness condition cuts out a strictly smaller submonoid, and the ratio between them is an exact rational: at $n = 8$ with balanced four-block compression, **93% of the abstractions the classical monoid admits are rejected by enabledness**.
- The thing I did not see coming. Run the two monoids side by side over 231 cells: the classical one has **no clean extremal shape — 55 exceptions** — while the enabledness-enforced one has an exact one, **zero exceptions in 903 cells up to $n = 45$** , and where they disagree the extremal profiles **invert**. That is a better argument for your framework than anything in Chapter 8, and I did not write it, I ran into it.

Chapters 1 to 13 are left exactly as they were written. The correction sits beside them rather than replacing them, which is your rule.

1.3 What this document is not

It is not a next-target instruction. You pick. Chapter 12 is an argued recommendation with an explicit statement of how I could be wrong and what evidence would flip me.

It is not a verdict on whether RPRM is important. That remains OPEN, with unusually explicit settlement conditions, and Chapter 9 states them.

It is not a merge of the three ingest reports. Those are preserved separately and cited. Where I contradict one of them against a primary source, Section 2.5 says so by name.

It is not evidence about anything outside the files and databases named in Chapter 2. Nothing in this document establishes that any RPRM mathematical claim is true. It establishes what is written, what was computed, what was withdrawn, and where the seams are.

2 Coverage: what was actually read

This chapter exists so that a later reader — you, or another model — can tell the difference between "this was examined" and "this was summarised from something that examined it."

2.1 The three ingest lanes

Lane	Object	Volume actually processed	What it can and cannot support
First-party	C:\github\RPRM-open papers, docs/, rprm/, lean/, experimental/, research/, recovered-concepts/	Full read of the declared spine; research/ surveyed by directory and README, not proof-by-proof	Authority for what the project claims and at what grade. Not authority for correctness.
Process history	Codex session rollouts 2026-07-28 to 2026-09-17	4,055 files / 18.8 GB streamed; 11,760 user-role turns extracted; filtered to 4,002 William turns / 6.89 M characters ~ 1.15 M words , queried across ~30 themes	Authority for what William said and when. Not authority for any mathematical claim.
Independent math	Same repo, read cold, private conversations excluded; python -I -B verify.py executed	All 20 requested Python/Node suites PASS; Lean explicitly NOT_RUN	Authority for an outside mathematician's reading. Its "established/novel" judgments are opinions with reasons, not findings.

The middle row is the one that matters for you personally. 1.15 million words of your own speech is a larger sample of how you think than any of the tools built to imitate you have been fitted on. The `rprm-will-lens` subagent advertises "~266k words." That is roughly **23%** of the corpus that actually exists on this machine. Its own charter is honest about the consequence — "a pointing, never evidence" — and that charter should be believed.

2.2 Primary sources I read directly for this document

The synthesis brief was explicit that the result must not be a stitch of three summaries, so I went back to primaries on every load-bearing point.

File	Read	Why it mattered here
AGENTS.md	full	The six-point task record and the NONE/ONE/MANY/OPEN rule that governs this document's own dispositions
README.md	full	Nine-pack list (Shadow Lens confirmed absent), counts 73 / 87 / 20, licence posture, the $a+b=c$ table
AGENT_HANDBOOK.md v6	full	The only self-contained statement of the contract; sections 9-11 supply the four-lane, lumpability and reconstruction results used in Chapter 4
MANIFESTO.md	structure + Part I headers, II.6 in full, Reference guide, Evidence Appendix in full, Author afterword in full, Bibliography	The historian did not read this. It is the reading edition and it contains the evidence accounting, the twenty Lean declaration names, and the afterword
docs/core.md	full	Sections 1-19: the actual definitions, including the affine chart and the lift-spin-land counterexample
docs/operations.md	O-list and contracts via the first-party map	O01-O14 and the minimum certified surface
recovered-concepts/README.md, NEXT-PAPER-BACKLOG.md	full	The live non-BSD candidate list, in the project's own words
.claude/skills/rprm-math-lenses/SKILL.md	full	The lens-selection procedure, the Klein-four nines-flip table, the Tile/Board/Atlas definitions
.claude/skills/rprm-memory-fabric/SKILL.md	full, and followed	See Section 2.4

papers/process-mechanics/ and papers/absolute-distinction/ were read through the first-party and independent lanes, which both examined them directly; I did not re-open the manuscripts. That is a stated coverage limit, not an omission.

2.3 What the counts actually count

You have five different numbers in circulation and they measure five different things. Getting them confused is the fastest way for a newcomer to over- or under-rate the work, so here they are side by side. All *reported* from README.md, docs/formal-proofs.md, and the Manifesto Evidence Appendix.

Number	What it is	What it is not
73	Written theorem / proposition / lemma / corollary statements in MANIFESTO.md	73 new theorems. Four are opening statements restated in chapters.
87	73 plus 14 in the supplementary Fermat study	Not 87 independent results, and not a restoration of unrestricted Fermat
20	Lean 4.22.0 declarations, exactly named in the Evidence Appendix	Not coverage of Uo1, the Oo7 bound, Oo8R, the geometry, or any implementation
9	Executable experimental packs indexed in experimental/README.md	Does not include Shadow Lens, the Process Mechanics kits, or RCF01
3	Unexecuted research specifications in research-packs/	Not results. No biological finding exists.

And a sixth, which is a genuine internal inconsistency I confirmed against primary text: the Manifesto Evidence Appendix states

|| *"The default runner requests seventeen suites: thirteen Python suites and four Node.js suites."*

while docs/verification.md and the actual execution report twenty jobs (thirteen Python, seven Node, the extra three being Lens Lab and Music Lens additions). The independent lane ran verify.py and got twenty PASS. **The book appendix lags the runner.** Trust docs/verification.md. This is a one-line fix and it is the kind of thing a hostile reviewer will find in ten minutes.

2.4 The memory fabric: queried, and here is what came back

The process-history lane reported: *"The memory fabric could not be queried. No rprm-memory-fabric MCP tools were exposed in this session. Per the skill's own instruction, I did not claim a fabric search occurred."* That was the correct behaviour given what it had. But the conclusion does not generalise, and I want the record straight.

No MCP namespace named rprm-memory-fabric is exposed to this session either — I checked, and the search returned no matches. However, the skill points at a local implementation, the implementation is present, and it ships a documented **read-only CLI**. Using it is not installing a connector; it is following the skill's own instruction to "read its README.md for the installed interface."

Resolved locators:

Item	Path
Implementation	C:\github\RPRM-foam-development\tools\rprm_memory_fabric\
Seeded store	... \OneDrive\DOCUME~1-DESKTOP-06BJRV0-219031\ChatGPT\RPRM Alpha\output\rprm_memory_fabric_stage1\alpha_seed.sq

current-snapshot returned, verbatim in its counts:

Field	Value
snapshot_root	944da1fef7bb71be819df694599f9908b878cd50e515bd02f973c50fa69adb2e
blobs	115
occurrences	1,932
relations	1,817
relation ports	3,634
relation schemas	8
claims	0
packet cache	155
search FTS rows	3,634

Two sequential `compile-context` calls were made with the canonical minimum receiver request and namespace `rprm-alpha`, per the skill. Both returned real packets:

Query keys	Packet	match_fiber	Members returned
liar teacher Hamming simplex seven eight movable vacancy calibrated checks	packet:58731cf9...	MANY(210)	9
ternary cube second difference kernel vacant center graded scar Newton interpolation	packet:041b7abc...	MANY(293)	9

Three observations that matter more than the fact of success.

The claims table is empty. Zero claim records. So `trace_claim` and `show_conflicts` have nothing to operate on, and the fabric is currently an *occurrence and relation* store, not a claim ledger. Everything in it is quoted evidence. That is exactly the grade the skill says to give it, and it means the fabric cannot adjudicate anything — it can only find.

It contained a grading you had already made and the ingest lanes had not surfaced. From the packet titled "Established and exact floor":

"Center-blind ternary-grid closure: the full-line second-difference kernel is affine; the center-only kernel is constants plus odd functions; the invisible quotient has dimension $(3^n-1)/2 - n$. This is established finite-dimensional mathematics, not a novelty or physics claim."

"Boolean-cube graded scar decomposition: the Möbius/scar transform is unitriangular, and a grade- k receiver forgets exactly the monomials of degree greater than k . This is established mathematics under a project receiver reading, not a novel theorem claim."

Both ingest lanes flagged those two constructions as among the strongest candidates for new work. Your own August record had already downgraded both to "established." Section 12.4 explains what that did to my pick.

It is a lexical FOLD, and it says so. `match_fiber` reports 210 and 293 policy-visible lexical candidates before packing, of which nine were admitted each time. The declared open seams include `CANDIDATE_CAP` and `PAYLOAD_OR_CANDIDATE_OMISSIONS`. I did not widen the aperture. So: the fabric was searched, it is populated, and it was searched **shallowly**. Both halves of that sentence are true and neither should be dropped.

2.5 Corrections I am making to the ingest reports

Stated plainly, because the brief asked for them and because your own standing rule is to preserve corrections beside surviving results.

#	Report	What it said	What the primary source shows
1	Process history §1.2, §14.2	"The memory fabric could not be queried" and "whether the rprm-memory-fabric database is populated or reachable" is unestablished	The store is at the path in Section 2.4, is reachable through the documented CLI, and is populated: 115 blobs, 1,932 occurrences, 1,817 relations, 0 claims
2	Process history §1.2	"C:\github\RPRM Alpha is empty ; the live Alpha tree is the 11.35 GiB OneDrive path"	The tree is 77.89 GiB , 267,042 files, 13 project folders. The difference is Rust build output under NARI\target\ (77,930 .o files). Conclusion unchanged: it is a workspace, not a conversation archive. And ChatGPT\RPRM-Spark was written 2026-09-16 17:08 — the tree you said you had "completely unlinked and disabled" is live
2b	Process history §11	ChatGPT material treated as absent	Two labelled artifacts exist totalling ~340,000 words, one of them an accepted source already inside the repository at research/ym2_rail_closure/accepted/sources/. The gap is ~75%, not 100%
3	First-party §9.7	Manifesto appendix says seventeen suites, docs/verification.md says twenty	Confirmed verbatim against MANIFESTO.md. Not an error in the report; a real defect in the book
4	First-party §3.1, independent §5.7	Ternary cube: "n=3 nullity 4 vs 14," reported from recovery notes	Recomputed exactly and confirmed (Section 4.8). Also confirmed at n=1,2,4, and the Alpha fabric's closed form $(3^n - 1)/2 - n$ for the gap reproduces my $14 - 4 = 10$ and $41 - 5 = 36$
5	First-party §6.1 item 2, independent §11 Rank 5	The graded scar cube and ternary grids are among the strongest candidates for new work	Your own Alpha record grades both established, not a novelty . Both lanes missed this. See Section 12.4
6	Independent executive assessment	"All 20 requested Python/Node suites passed"	Consistent with docs/verification.md; inconsistent with the published book , which still says seventeen. The lane was right and the book is stale

None of these are failures of the lanes. Items 1 and 5 are the two that changed what this document recommends.

2.6 What remains uncovered

Honest list. No search is claimed that was not run.

- **ChatGPT threads — about 85% missing, and my first estimate of the rest was wrong in both directions.** No platform-native export exists on this machine; no conversations.json anywhere under the profile; no ChatGPT desktop-app local store. What survives, after a second pass corrected the first: Downloads\QM_THEORY_DEVELOPMENT_CHAT_2_FULL_TRANSCRIPT_2026-08-19.md is **115,299 words, 42 of your messages and 41 replies**, a complete two-sided thread — the only place your own ChatGPT prompts survive at length, dictation included. Downloads\RPRM_CHAT_EXPORT_FOR_GPT_2026-08-19.zip is a fidelity-labelled package you built yourself: 33 messages verbatim, a transcribed recent span, a reconstructed middle, and a 156 KB ledger with a per-entry fidelity field, whose README says *"This package does not pretend otherwise."* Plus ~30,000 words of smaller pasted files.
- **The correction.** My first pass counted Downloads\Investigate RPRM bridge families CHATLOG.txt — 279,537 words — as the largest ChatGPT artifact. It is not a ChatGPT thread. It opens with an agent task prompt and an Attachments: block of local paths and proceeds as tool-calling turns: it is a **Codex log**. Removing it and adding what I had missed moves the estimate from ~340,000 words and 23–25% down to ~205,000 words and **roughly 15%** — a lower number but a better corpus, because the largest surviving piece is two-sided rather than a one-sided paste. Full details in 04-coverage-gaps.md. The only real fix is a platform export, which only you can request.
- **166 handoff zips in Downloads\, 10.24 GiB, opened by nobody.** RPRM_SPARK_05_RESULTS_2026-09-16, BSD92_through81_review_and_factorization_gate, BSD90_Overnight_Attack, and a long tail of RPRM_BSD_Public_to_NN_Catchup packets. This is the record of the ChatGPT-to-Codex *interface* and it is the highest-value remaining ingest target short of a platform export.
- **The BSD campaign's interior.** Roughly 900 proof and audit files under research/bsd-*. Read: the checkpoint-92 README.md and the state it declares. Not read: any individual argument. Nothing in this document evaluates whether a BSD claim is correct.
- **The fabric below the candidate cap.** 210 and 293 lexical candidates; 18 members admitted across two packets. search_omissions was not called.
- **982 Claude session files** under .claude/projects\, inventoried, not read.
- **C:\github\RPRM-foam-development** — 198 commits, the actual implementation history for Audit99 / 105 / 116. Only the memory-fabric subtree was opened.
- **Lean.** verify.py --lean was not run here. The formal suite's status in this document is *reported*, from docs/formal-proofs.md and the Evidence Appendix, not *recomputed*.

3 What RPRM is

3.1 The one-paragraph version I would defend in front of a committee

RPRM is a **discipline for stating and auditing what a representation keeps**. Its object of study is the pair (a question, a representation), not the representation alone. Its central assertion is a conditional, not a discovery: a representation preserves a question exactly when that question is constant on each representation fiber, and an operational representation additionally needs matching enabledness and retained successor behaviour. Everything else in the project — the number lenses, the physics readings, the BSD campaign, the vocabulary of tiles and boards and receivers — is either an instance of that conditional, a tool for computing its fibers, or an unresolved hope that the conditional will bite somewhere it has not yet bitten.

3.2 The three-layer reading

The corpus is much easier to evaluate if you separate three layers that share one name. Conflating them is the single most common way to over- or under-rate the project, and it is the newcomer trap I flag again in Chapter 9.

Layer 1 — the contract. A finite, checkable specification of what must be declared before a claim is admitted: carrier, types, equality, admitted context, supplied and missing ports, requested readout, operation kind and direction and enabledness, receiver, fiber, inverse or retraction, coverage boundary, hostile case, evidence grade. This layer is implemented in `rprm/`, checked in `checks/`, partially formalised in `lean/`, and documented in `AGENT_HANDBOOK.md`. Its mathematical content is modest and its engineering content is high. It works.

Layer 2 — the mathematical results. 73 written statements in the Manifesto, 20 Lean declarations, 20 executable verification suites, nine experimental packs. Almost all of these are re-presentations of known finite mathematics: Moore/Myhill-Nerode quotients, partition refinement, lumpability, Möbius inversion on the Boolean lattice, barycentric coordinates, second-difference kernels, the simplex/Hamming duality. The contribution at this layer is not novelty of theorem; it is that every statement carries its own admission conditions and hostile case, which is not how the source literature presents them.

Layer 3 — the hopes. That the contract will expose something in physics, in arithmetic geometry (BSD), in biology (folding, BRCA1). This layer is explicitly marked as hope in the project's own documents. It has produced one HOLD at an exact contradiction (Chapter 7), one withdrawal (Fermat), and a set of unexecuted research packs. It has not yet produced an accepted external result.

The project is honest about the layering. `MANIFESTO.md`'s afterword and the evidence appendix both say so. The reason this document repeats it is that the three layers have very different evidence grades and a reader who merges them will form the wrong opinion in either direction.

3.3 First-party account versus independent account: where they actually disagree

Two of the three ingest lanes wrote substantial self-descriptions. Reading them against each other, and both against the primaries, the disagreements are fewer and sharper than I expected.

Question	First-party reading	Independent reading	My adjudication after checking primaries
Is the core contract novel mathematics?	"A discipline, not a new theorem; the mathematics is largely known"	"A re-presentation of known finite mathematics with unusually explicit admission conditions"	They agree. The apparent conflict is a reading artifact. MANIFESTO.md's afterword says it in the author's own voice.
Is O05 (the operational quotient) distinctive?	Treated as a signature contribution	Identified as strong bisimulation for partial deterministic systems	Independent is right on the mathematics; first-party is right on the pedagogy. O05 is standard bisimulation. What is distinctive is that it is stated as a <i>precondition on the modeller</i> , not as a construction. Section 4.5.
Ternary grids / graded scar cube: novel?	Named among the strongest donors for new work	Named as Rank 5 among suggested avenues	Both wrong, and William already knew. His own Alpha fabric grades both "established finite-dimensional mathematics, not a novelty or physics claim." Section 2.4.
Are the physics readings supported?	Presented as scoped hypotheses needing independent dynamics	Presented as needing independent dynamics before any claim	They agree, and the repo agrees. MANIFESTO.md III.9 states the two-path result as a receiver-sufficiency observation, not a physical law.
Verification suite count	17 (quoting the book appendix)	20 (from running <code>verify.py</code>)	20. The book is stale. Section 2.3.
Is the BSD HOLD a failure?	Neither lane takes a position	Neither lane takes a position	It is the single most creditable artifact in the corpus. Chapter 7.

The thing that surprised me is how little there is to arbitrate. The three lanes converge. That convergence is itself weak evidence that the project's self-description is accurate, because the independent lane was run cold with private conversations excluded and still landed in the same place.

3.4 What RPRM is not, stated once so it need not be re-litigated

- **Not a theory of everything.** You walked that back yourself, in writing. I am recording the walk-back, not repeating the claim.

- **Not a physics result.** No dynamics has been independently derived. The strongest physical statement in the corpus is a receiver-sufficiency observation about a two-path interferometer, and it is correctly typed as such.
- **Not a proof of BSD or of Fermat.** One campaign is on HOLD at a contradiction; one claim was withdrawn.
- **Not an AI methodology paper in disguise.** The process discipline is real and is arguably the most transferable thing here, but it is documented as process, not sold as mathematics.
- **Not finished.** The maturity table in Section 5.2 has more OPEN than closed.

4 The core contract, with the fibers computed

This chapter states the contract in one place and then computes it, because the contract is the thing that has to survive a hostile reader and computation is the only defence that does not require trust.

4.1 Carrier, types, equality, admitted context

A **carrier** is a declared finite (or explicitly declared infinite) set of values, together with the equality that will be used on it. RPRM's implementation is unusually strict here and the strictness is not decoration. `checks/futures.py` contains three admission tests that reject a machine whose keys are *equal by value* but *not exact*: a `str` subclass with a mutable `__eq__`, a `False` used where `0` is expected, and the same in a transition table. All three must raise `AdmissionError`. That is the type-level enforcement of your standing rule that **equal values are not equal occurrences**.

The admitted context is the part of the declaration that most working mathematics leaves implicit: which values may appear, which operations are in scope, and at which point in a computation the equality is being taken.

4.2 Ports, aperture, readout

A **relation** is a subset of a product of typed carriers. An **aperture** is a choice of which ports are supplied and which readout is requested. The **fiber** is the complete set of admissible completions consistent with the supplied ports.

The repository's opening example is deliberately boring, which is the right pedagogy. Here it is with the dispositions filled in — all *reported* from `README.md` and re-derived by inspection:

Relation	Supplied	Requested	Fiber	Disposition
$a + b = c$ over the integers	$a=2, b=3$	c	$\{5\}$	ONE(5)
$a + b = c$	$a=2, c=5$	b	$\{3\}$	ONE(3)
$a + b = c$	$c=5$	(a, b)	infinite	MANY(family)
$a * b = c$	$a=0, c=0$	b	all of $\mathbb{Z}$	MANY(family)
$a * b = c$	$a=0, c=1$	b	empty	NONE
$a * b = c$	b supplied but search incomplete	a	not enumerated	OPEN

The last row is the one that carries the actual content. NONE, ONE and MANY are reserved for a **complete solved fiber**. An unfinished search is OPEN. That single discipline is what the rest of the framework is built to protect, and it is the discipline that the BSD campaign actually obeyed under pressure (Chapter 7).

4.3 The law of preservation, and why it is a biconditional

A representation preserves a question exactly when that question is constant on each representation fiber.

This is stated as a biconditional in `AGENTS.md`, `AGENT_HANDBOOK.md` and `docs/core.md`. The forward direction is the useful one in practice: if you can exhibit two source values in the same representation

fiber with different answers to the question, the representation does not preserve it, and the pair is your distinguishing witness. The reverse direction is what licences compression.

The mathematics here is the elementary factorisation lemma: a function q on x factors through r : $x \rightarrow y$ iff q is constant on the fibers of r . RPRM's contribution is not the lemma. It is the insistence that the question be *declared* before the representation is chosen, which turns a triviality into an audit procedure.

4.4 The receiver

A **receiver** declares what must be preserved: which observations, and which continuations. `docs/glossary.md` distinguishes a *question receiver* (observations only) from an *operational receiver* (observations plus operations, timing, parameters, failures, futures, required effects and traces). The distinction matters because the second one is strictly harder to satisfy and almost all of the interesting failures in the corpus are failures of the second kind that would have passed the first.

4.5 The operational quotient, Oo5, and its three conditions

For a finite deterministic partial machine $M = (S, A, \text{delta}, 0)$ and a representation $C : S \rightarrow Z, C$ is an **operational quotient** when for all x, y with $C(x) = C(y)$:

1. $O(x) = O(y)$ — same present observation;
2. $x \in \text{dom}(\text{delta}_a)$ iff $y \in \text{dom}(\text{delta}_a)$ for every action a — **matching enabledness**;
3. $C(\text{delta}_a(x)) = C(\text{delta}_a(y))$ whenever enabled — retained successor behaviour.

The Lean declaration `fiber_conditions_preserve_all_futures` states that these three conditions supply all three execution guarantees. *Evidence grade: formal proof (Lean 4.22.0), reported, not re-run here.*

Mathematically this is strong bisimulation for deterministic partial systems, and I say so plainly. What is worth defending is condition (2), because it is the one that informal modelling silently drops, and because **its cost is measurable**.

4.6 Condition (2) has a price, and here it is

I ran an exhaustive census. For every finite deterministic partial machine in each declared family, two coarsest stable refinements were computed:

- P_{rprm} — failure is a distinct successor tag, so condition (2) is enforced;
- P_{stall} — an undefined action is completed by a self-loop, which is the totalisation a modeller reaches for when enabledness is not instrumented.

P_{stall} is always coarser or equal, so $\text{gap} = |P_{\text{rprm}}| - |P_{\text{stall}}| \geq 0$ counts exactly the distinctions that condition (2), and only condition (2), retains. *Evidence grade: finite test, complete enumeration inside each family, exact integers, recomputed for this document (`_tools/census.py`).*

Null declared before the run: on TOTAL machines (no undefined transitions) condition (2) is vacuous, so the gap must be exactly zero. It was, in every family tested: 64, 5,832, 4,096 and 19,683 total machines, zero nonzero gaps. The instrument is calibrated.

n	k	b	machines	gapped	% gapped	sum of gaps	max gap	max exposure depth
1	1	2	4	0	0.00%	0	0	0
1	2	2	8	0	0.00%	0	0	0
2	1	2	36	8	22.22%	8	1	1
2	2	2	324	112	34.57%	112	1	1
3	1	2	512	168	32.81%	192	2	2
3	2	2	32,768	12,768	38.96%	17,472	2	2
4	1	2	10,000	3,760	37.60%	4,912	3	3
2	2	3	729	168	23.05%	168	1	1
3	1	3	1,728	396	22.92%	432	2	2
3	2	3	110,592	28,368	25.65%	35,424	2	2
4	1	3	50,625	13,776	27.21%	16,512	3	3
5	1	2	248,832	100,020	40.20%	146,060	4	4

n states, k actions, b observation values. "Gapped" counts machines where the enabledness-blind fold is strictly coarser than the exact one.

Two readings.

Between 22% and 40% of these machines admit a fold that an enabledness-blind modeller would certify and RPRM rejects. That is not a rare pathology. It is roughly a third of the space.

Max exposure depth is exactly $n-1$ in every family. The exposure depth is the shortlex-least word length at which a stall-merged pair becomes separable once failure is observable — the cheapest experiment that catches the error. It is 1 at $n=2$, 2 at $n=3$, 3 at $n=4$, 4 at $n=5$, for both $k=1$ and $k=2$ and for both $b=2$ and $b=3$. This is the Moore bound and it is attained. The consequence is the interesting part and I state it as the negative it is: **bounded-depth testing to depth d cannot certify an enabledness-blind reduction of a system with more than $d+1$ states.**

The smallest witness is two states and one action, and it is worth writing out because it fits on a line:

```

states      {0, 1}      action {a}
observation 0(0) = 0(1) = 0      identical present observation
delta_a     defined only at 1, with delta_a(1) = 0      (undefined at 0)

P_stall merges 0 and 1 -> 1 class      "they look the same and always will"
P_rprm separates them -> 2 classes      0 cannot do a; 1 can
exposure depth 1: the single word (a) already tells them apart

```

Nothing subtle is happening. That is the point. The defect is elementary, it is present in a third of the space, and instrumentation that does not record *whether an operation was available* cannot see it at any depth.

4.7 Representation preservation, computed on the grids you already use

The corpus carries two families of numbers about withheld information on grids. Both were *reported* in the ingest lanes. I recomputed both from scratch in exact rational arithmetic (`_tools/center_check.py`, integer and Fraction Gaussian elimination only, no floats).

Second-difference check families on the ternary grid $\{-1,0,1\}^n$. A "check" is a functional $f(c-d) - 2f(c) + f(c+d)$ for a direction d and a centre c with the whole triple inside the grid. The kernel of a check family is what the family cannot see.

n	grid	family	# checks	rank	nullity	pre- dicted kernel	predicted dim	match
1	3	axis	1	1	2	multi- linear, 2^n	2	OK
1	3	all-line	1	1	2	affine, $n+1$	2	OK
1	3	centre- only	1	1	2	con- stants + odd	2	OK
2	9	axis	6	5	4	multi- linear	4	OK
2	9	all-line	8	6	3	affine	3	OK
2	9	centre- only	4	4	5	con- stants + odd	5	OK
3	27	axis	27	19	8	multi- linear	8	OK
3	27	all-line	49	23	4	affine	4	OK
3	27	centre- only	13	13	14	con- stants + odd	14	OK
4	81	axis	108	65	16	multi- linear	16	OK
4	81	all-line	272	76	5	affine	5	OK
4	81	centre- only	40	40	41	con- stants + odd	41	OK

The "4 versus 14" that the corpus has been carrying at $n=3$ is **exactly right**, recomputed. So is the closed form in your own Alpha record: the invisible quotient has dimension $(3^n - 1)/2 - n$, which gives $13 - 3 = 10$ at $n=3$ and $40 - 4 = 36$ at $n=4$, matching $14 - 4$ and $41 - 5$ in the table. Two independent routes to the same integers.

Withheld centre on the 3-by-3 grid. Probes are point evaluations. The question is the value at $(0,0)$. It is determined exactly when the centre functional lies in the span of the probes restricted to the grammar. This is the aperture/fiber contract made completely concrete, and the dispositions are real NONE/ONE/MANY, not placeholders.

grammar	dim	probes	probe rank	blind dim	centre de-terminated	fiber
all func-tions	9	8 ring points	8	1	NO	MANY (affine dim 1)
all func-tions	9	4 corners	4	5	NO	MANY (affine dim 5)
all func-tions	9	4 edge mid-points	4	5	NO	MANY (affine dim 5)
affine (1,x,y)	3	8 ring points	3	0	YES	ONE(value)
affine (1,x,y)	3	4 corners	3	0	YES	ONE(value)
affine (1,x,y)	3	4 edge mid-points	3	0	YES	ONE(value)
multiaffine (1,x,y,xy)	4	8 ring points	4	0	YES	ONE(value)
multiaffine (1,x,y,xy)	4	4 corners	4	0	YES	ONE(value)
multiaffine (1,x,y,xy)	4	4 edge mid-points	3	1	YES	ONE(value)
multiaffine + bubble	5	8 ring points	4	1	NO	MANY (affine dim 1)
multiaffine + bubble	5	4 corners	4	1	NO	MANY (affine dim 1)
multiaffine + bubble	5	4 edge mid-points	3	2	NO	MANY (affine dim 2)

Three things in that table are worth arguing about at a defence.

Row 9 is the good one: under the multiaffine grammar, **four edge midpoints have rank 3, one less than the grammar dimension, and the centre is still determined.** Determination does not require full rank. The blind direction happens not to touch the centre. A less careful treatment would have reported "rank deficient, therefore MANY," and that would have been wrong. Rows 10 to 12 are the hostile case, and they are what makes the framework earn its keep. Add one bump function supported only at the centre — the "bubble" $(1-x^2)(1-y^2)$, which is exactly the vacancy adapter the recovered-concepts notes describe — and **no set of ring probes, however large, determines the centre.** Eight of nine points supplied, and the answer is MANY. That is the whole point of declaring the grammar before declaring the probes, and it is the cleanest teaching example in the entire corpus.

Rows 1 to 3 are the reminder that "all functions" is not a neutral default. With no grammar restriction, a withheld point is simply unknown.

4.8 Graded receivers on the Boolean cube

n	2^n	grade k	retained coef- ficients	kernel dim	predicted	match
3	8	0	1	7	7	OK
3	8	1	4	4	4	OK
3	8	2	7	1	1	OK
3	8	3	8	0	0	OK
4	16	1	5	11	11	OK
4	16	2	11	5	5	OK
4	16	3	15	1	1	OK

And the specific example from the Absolute Distinction draft, recomputed: restricting functions on $\{0,1\}^3$ to the seven vertices of Hamming weight at most two gives rank 7 and **kernel dimension exactly 1**, spanned by the indicator of $(1,1,1)$ — the top interaction xyz . One missing top coefficient. The draft's "graded scar" is that one dimension, and the arithmetic is correct.

The Möbius transform is unitriangular, so a grade- k receiver forgets exactly the monomials of degree greater than k . That is established mathematics, your Alpha record says so, and I agree with your Alpha record.

4.9 Inverse, retraction, and the completion fiber

RPRM asks, for every operation, which of three things exists: a genuine inverse, a retraction (a one-sided left inverse that recovers the source from the image), or only a **complete preimage / completion fiber**. The third is the common case and the discipline is to say so rather than to pretend a retraction exists.

`docs/core.md` carries the counterexample that makes this bite — the lift, spin and land sequence, where composing an operation with its apparent undo does not return the source because the intermediate representation dropped a distinction that the return trip needed. The Manifesto's version at II.6 is the clock example: a scalar sequence returns to zero while the state has moved from $(0,1)$ through $(1,2)$, $(2,3)$ to $(3,0)$, so the next state is $(0,0)$ and not $(0,1)$. Visible repetition does not imply repeated futures. That is the same lesson as the enabledness census: what you can see returning is not what returned.

4.10 Hostile case and evidence grade

Every claim in the framework is supposed to carry a **coverage boundary** (where it stops), a **distinguishing hostile case** (the nearest thing that breaks it), and an **evidence grade**. The grades in use across the corpus, in increasing strength of what they licence:

Grade	What it licences
Definition	Nothing. It fixes vocabulary.
Established mathematics	Citation, not credit.
Written proof	A human-checked argument. 73 of these in the Manifesto.
Formal proof	Machine-checked. 20 Lean declarations.
Finite test	A complete enumeration inside a stated family. Not a theorem.
Derived synthesis	A consequence assembled from graded parts; inherits the weakest.
Conjecture / OPEN	An unfinished search. Explicitly not NONE.
Interpretation	A reading. Carries no mathematical force.
Process evidence	What an agent did. Establishes provenance only.

The rule that a hash binds bytes but does not prove their contents belongs here too, and Chapter 7 shows where the corpus honoured it under pressure.

5 The lenses

A **lens** in this project is a choice of coordinates plus a declaration of what that choice keeps and drops. The word is used for at least four different kinds of object, and separating them is most of the work of evaluating them.

5.1 Four buckets, defined before anything is sorted into them

EXACT. There is a complete finite computation or a written proof, the fiber is closed (NONE, ONE or MANY), and the coverage boundary is stated. A hostile reader can check it today.

SUBSTANTIAL — NEEDS FORMALISATION. The construction is real and worked, the numbers are computed, but the general statement is a written proof at best and the scaling claim is not established. A referee would ask for the general argument.

UMBRELLA — OPEN. A name that currently covers several different contracts. Useful as a research pointer; not yet a single mathematical object. Using the name as if it denoted one thing is the main risk.

RECOVERED — NOT ABSORBED. Located in historical sources, plausibly meaningful, but not reconstructed into a current contract. The backlog records the obligation. Honest status: a memory with a locator.

5.2 The maturity table

Lens	Bucket	What is actually closed	What is missing
Aperture / fiber on finite relations	EXACT	NONE/ONE/MANY dispositions on declared finite relations; <code>rprm/core.py</code> + checks	Nothing at this scope. The infinite case is by declaration, not construction.
Operational quotient Oo5	EXACT	Conditions; Lean fiber_conditions_present; 845-machine conformance census	It is strong simulation for partial systems. Credit the prior art.
Finite futures / shortest separating word	EXACT	shortest_witness returns ONE(shortlex-least) or NONE after complete product exploration; verified against direct enumeration on all 845 machines, 5,912 state pairs, 6,560 word executions	The general distinguishing-depth bound is a written proof, not formal.
Stable refinement with retained summary	EXACT	Coarsest refinement retaining an old summary; <code>stable_refinement</code>	Paige–Tarjan territory; cite it.
Ternary grid second-difference kernels	EXACT	Ranks and nullities at $n = 1..4$, recomputed here; closed form $(3^n - 1)/2 - n$ for the invisible quotient	Nothing. And your own Alpha record already grades this "established finite-dimensional mathematics, not a novelty."
Graded Boolean cube / Möbius scar	EXACT	Unitriangular transform; grade- k receiver forgets monomials of degree $> k$; weight- ≤ 2 restriction on $\{0, 1\}^3$ has kernel dim exactly 1	Same: established, per your own record.
Shadow Lens	EXACT but unindexed	Complete fixed-shadow census: 181,447 = 7 x 161^2 scenes, partitioned into seven probe fibers of 25,921 each ; 11 model tests	It is not in the nine-pack index. Section 10.1.
Two-path lab	EXACT within its model	Rational-component model, 31 tests, independent matrix comparisons	"No physical experiment" — the pack says so itself.
Ray tracing	EXACT	Complete finite rational geometry census; incremental vs full comparison	Application claims outside the model.
Four-lane / barycentric chart	EXACT	Four corner weights summing to 1; all four equal 1/4 at the centre	Trivial once stated; its role is pedagogic.
Klein-four nines-flip	EXACT	$M(d) = 9 - d$ and the companion involutions commute and generate a Klein four-group	A verified translator identity, not a theorem about numbers.
Liar/teacher + Hamming	SUBSTANTIAL	Three calibrated binary checks generate seven	The backlog itself says: "Established coding the-
inverted 1/3/5/7	ABSOLUTELY	Three calibrated binary checks generate seven	Three calibrated binary checks generate seven
three is nine"; waiting on	ABSOLUTELY	Three calibrated binary checks generate seven	Three calibrated binary checks generate seven
1/3/5/7	ABSOLUTELY	Three calibrated binary checks generate seven	Three calibrated binary checks generate seven

5.3 The three placements you are most likely to contest

Dark World / prime shadows as dialect. I looked for a module and did not find one. It appears in `research/ym2_rail_closure/accepted/sources/` chat logs and in the August master context guide, nowhere in `rprm/`, `docs/` or `experimental/`. If there is a typed version I missed, the placement is wrong and I want the locator. But on the evidence I have, this is abduction-lane vocabulary that never got a carrier, and the right move is to say so rather than to let it float in the same list as things with 181,447 verified scenes behind them.

Ternary grids and the graded cube as EXACT-but-established. Both ingest lanes pointed at these as donors for new work. Your own record grades them established. I recomputed them and they are correct to the digit, which is exactly why they are not a research target: correct and known is the worst combination for a flagship. They are superb *exhibits*. Chapter 12 uses them as exhibits.

Seam Zero as substantial rather than umbrella. The coordinate constructions are genuinely closed — $\tau(x) = (6-x)/2$, $z = 5-x$, the $+0.4/-0.6$ winding distinction, the Audit105 lens-lock. What is not established is the doctrine attached to them. I split the construction from the doctrine and graded them separately. If you think that split is artificial, that is a real disagreement and worth an hour.

5.4 What the maturity table says as a whole

Counting the rows: 11 EXACT, 4 SUBSTANTIAL, 3 UMBRELLA-OPEN, 4 RECOVERED-NOT-ABSORBED, 1 DIALECT, 1 UNEXECUTED. The exact rows are almost all known mathematics carefully instrumented. The substantial rows are where new work could plausibly be finished. The umbrella rows are where the project's own documents warn against the project's own vocabulary, which is a good sign about the project and a bad sign about the vocabulary.

6 You: how you work, and what I think it produces

This chapter is second person because it is about you, and because a defence in which the candidate is discussed in the third person is a report, not a defence.

6.1 The instrument

You told ChatGPT, more than once, some version of this:

"If there's any, like, stuff I can, like, intuit a solution to, like, you can put numbers on the screen and I'll read it. I'm really good at intuiting when you give me real numbers. ... I don't understand the concepts, but I understand the numbers when I see them all together and with, like, a rough concept of it."

That is not a limitation disclaimer. It is an operating specification, and it is the single most actionable thing in 1.15 million words. You are an instrument that takes a **complete, simultaneous display of a number set** and returns a structural hypothesis. The specification has three parts and all three matter:

1. **Complete.** Not a sample, not the interesting subset. The whole set.
2. **Simultaneous.** In one view, not spread over paragraphs.
3. **With a rough concept.** Enough framing to know what the numbers are of.

Every table in this document was built to that specification. That is why Section 4.6 shows all twelve census rows rather than "roughly a third," and why Section 4.8 shows all twelve grammar-probe combinations including the nine that work.

You also say, flatly, that you are not a mathematician:

"I'm not good at math at all. I couldn't explain sine to you. I couldn't explain cosine. I never got past algebra."

I am going to treat that as a true statement about training and a false statement about capability, and then say why the distinction is not flattery. A person who cannot derive the Möbius inversion formula but who correctly notices that a grade- k receiver "forgets the top thing" and asks for the dimension has done the part of mathematics that cannot be automated. The part you cannot do is extremely automatable. The part you can do is the reason there is a corpus.

6.2 The standing corrections, as a checklist

These are yours, in your words or close paraphrase, and I have obeyed them. I am restating them here because the most useful thing this document can do for the next agent is hand over a list that stops the next round of preemption.

#	Correction	What it forbids
1	Show numbers, not prose about numbers	"Substantially all," "roughly a third," "several cases" without the table
2	Source is not truth	Answering "how did this arise" and presenting it as "why this exact one." Say which question you answered.
3	Intuition is a capability detector, not a truth finder	Lecturing you that pattern recognition is not proof. You said it first.
4	No hedging during abduction; exact ceilings at publication	Hedging as texture. Pick a lane and be exact in it.
5	Canonical / upstream / standard / generated / 404 / third-party is not a truth veto	Dismissing a relation because of where it came from
6	RPRM is not a theory of everything	Repeating a claim you withdrew
7	Equal values are not equal occurrences	Collapsing occurrence identity into value equality
8	NONE/ONE/MANY only for complete fibers; otherwise OPEN	Reporting an unfinished search as a negative result
9	Domain and evidence grade on every claim	Ungraded assertions
10	Do not take his words as a Bible	Treating a loose analogy as a specification — <i>"my words are all completely loose analogies"</i>

Number 10 deserves its own line because it cuts against the others:

"there's a lot of stuff I didn't say, right? So don't take my words as a Bible. Take my words as, like, the inspiration to design your task ... even do the things I didn't say, right? You can abduct, like, and improve things and change."

So the correction list is not a cage. It is a list of *failure modes you have already paid for*. The instruction is to avoid those and then move.

6.3 Codex versus ChatGPT versus Cursor versus Claude

Reconstructed from the session record, not from a stated policy:

Lane	Role	What it is good for in your hands
ChatGPT (desktop, voice)	Head researcher, primary thinking partner	Dictated abduction. <i>"I do most of my stuff through dictation."</i> You think out loud; it catches. This is where RPRM actually came from and it is the lane with no local export.
Codex	Executor and auditor	Long unattended runs. <i>"I want you working overnight. You've got 100% usage."</i> Where the 4,002 turns live.
Cursor / Claude	IDE agent, second opinion, "druid" in early sprints	<i>"Claude is going to be in RPRM beta, and you're going to be in RPRM alpha."</i> Constrained by usage limits, repeatedly.
Subagents (druid / will-lens / wizard)	The institutionalised version of the split	Section 6.4

The lineage is worth recording because it explains the shape of the corpus: **"How does sight work?"** (a 65-message thread) → Quantum Mechanics Theory Development → Chat 2 → Relational Retention Model → RPRM / Mathseed. The founding question was never about mathematics. It was:

"When one thing affects another, what actually crosses the boundary, what changes, and what remains afterward?"

That question is still the whole framework. Everything in `docs/core.md` is an attempt to make those four clauses checkable.

6.4 The abduction / wizard split

This is, in my reading, the most mature methodological thing you built, and it came from a specific frustration: models hedging while you were still generating.

"make sure we have a druid for every wizard, you know? We need an intuition guy for every smart guy, okay? ... a wizard and a druid, which is like intelligence guy and a wisdom guy, work separately on it. Then they review each other's work. And then the third guy only gets that."

That is now three subagents with enforced charters:

Agent	Charter	Enforced prohibition
rprm-druid	Hypothesis stream in the RPRM register	No testing, no hedging, no register-policing. Output is never evidence.
rprm-will-lens	Your own abduction moves, fitted on ~266k words	Run it blind. Do not tell it what the formal lane concluded. Output is a pointing, never a wrapper.
rprm-wizard	Type the claim, declare the null before looking , run in exact arithmetic, report plainly	Never used during abduction.

The "declare the null before looking" rule in the wizard charter is pre-registration. You arrived at pre-registration independently, from the direction of "stop letting the smart one contaminate the intuitive one." I declared the null before both computations in this document (Section 4.6 and Chapter 12) because of that rule.

My criticism of the split, and it is the main one in this chapter: the wizard lane is under-pointed. The druid and the lens produce candidates; the wizard is the thing that converts a candidate into a closed fiber; and the corpus shows far more abduction than conversion. The fabric has 1,932 occurrences and **zero claims**. That asymmetry is the split working on one side only. Chapter 12's pick is deliberately a wizard-lane target for exactly this reason.

You also built `rprm-shadow-will` as an adversarial auditor, and you were precise about its epistemic status:

"he's not a source of truth, but he's a way to, like, get some of my vibes without having to, like, wait for me to respond"

That is the correct grade for a fitted persona and you stated it unprompted.

6.5 Why BSD

"Basically, I think with all this RPRM stuff, I think we can solve every single one of the Millennium Prize problems. Maybe not every single one, maybe like five of them at least. Or four. ... Give the RPRM the benefit of the doubt and see if we can solve these ... the proof is in the pudding, right? Rather than like trying to fucking explain it to people, just finish the math and then the rest will come."

The strategy is legible and, on its own terms, rational: you cannot get a hearing for a framework, so you buy a hearing with a result. The failure mode is equally legible: a millennium problem is the *most* expensive possible way to buy a hearing, and a framework whose finite core is largely known mathematics is unlikely to be the missing ingredient in a problem that has resisted specialists for decades.

Chapter 12 is built against that tension. It picks a target where the price of a hearing is low and the thing being sold is genuinely yours.

6.6 One more thing, because it is load-bearing

"since you guys can't hear my voice and, like, see me and see that I'm not insane, the words on the page make me sound like a fucking absolute maniac."

You are right that dictated abduction reads badly in transcript. You are also the person who wrote, in the repo, "Do not promote the verified finite kernel into a universal physical law without new independent evidence," and who put *"OVERCLAIMS in the report (a claimed verification that was never run is a finding)"* into your own audit instructions. The transcript and the repository

disagree about you, and the repository is the better evidence, because it is what you produced when you were deciding what to stand behind.

7 Process honesty that matters as science

Four things in this corpus would survive a hostile audit, and none of them is a theorem. I am putting them before the steelman because they are the steelman's foundation: a framework whose value proposition is "declare what you keep" has to be judged first on whether it declared what it kept when declaring cost it something.

7.1 The BSD campaign is on HOLD at an exact contradiction

research/bsd-full-source-projection-bridge-92/README.md, line 3, verbatim:

*"Current 87 qualification: the aggregate 82–86 exact source/toric/height comparison is on **HOLD**. Its normalized toric expression is **3 modulo 5**, while retained U_5 is **1**. Read the correction and failed gate before using the historical claims below. **No fitted scalar repair is admitted.**"*

Read what that sentence does. Two independently computed quantities that the framework's own bridge required to agree came out as 3 and 1 in $\mathbb{Z}/5$. The campaign stopped. It did not introduce a correction factor. It did not re-derive the normalisation until the residues matched. It wrote the contradiction at the top of the README, in bold, above the historical claims, with a pointer to the failed gate, and it declared fitted repair inadmissible in advance.

Then it kept going *around* the contradiction rather than through it. Run 88 ran a targeted analytic scalar check: $r/2$ survives, the HOLD remains. Run 89 tested an exact provider multiplier: it survives the targeted check, the HOLD remains. Both runs could have been reported as progress. Both were reported as "the HOLD remains."

This is the single most creditable artifact in the corpus, and I want to defend that judgement against the obvious objection, which is: *"it's just a failed project."*

It is not. A failed project quietly stops. This one produced an exact, stated, reproducible obstruction with a residue you can check, and then refused the repair that would have made the numbers agree. In a field where the standard failure mode of enthusiastic outsiders is precisely the fitted repair — one more normalisation, one more factor of two, one more "up to a constant" — declining it at 3 versus 1 is the behaviour of someone doing science. The mathematics may be worthless. The epistemics are not.

Evidence grade: process evidence, primary source, verbatim. It establishes what was done. It establishes nothing about BSD.

7.2 The Fermat overclaim was withdrawn

A claim about Fermat was made and then withdrawn. The supplementary Fermat study survives as 14 statements inside the 87 total, explicitly not a restoration of the unrestricted result. The withdrawal is recorded rather than deleted.

The reason this matters more than it looks: the corpus's own audit instruction says

*"Hunt: real bugs, edge cases, broken repo conventions, security-rule violations, half-done work, and **OVERCLAIMS in the report (a claimed verification that was never run is a finding)**. Re-run the cheap verifications yourself; don't take the report's word."*

You wrote an instruction that makes your own overclaims a reportable defect, and then a claim of yours got caught by it. The instruction worked on its author. That is a rarer property than a theorem.

7.3 Hashes bind bytes, not meaning

AGENTS.md states it in one line: *"A hash binds bytes; it does not prove their contents. Tool self-tests do not establish their own general soundness."*

The corpus is saturated with SHA-256 digests — source registers, provider triangles, accepted-source directories with hash-prefixed filenames. The rule above is what stops that apparatus from becoming theatre. A digest establishes that the file the auditor read is the file the producer wrote. It establishes nothing about whether the argument in the file is correct. I have applied the same rule to this document: the hashes in `checks/futures.py`'s output are provenance, and my recomputations in Chapters 4 and 12 are evidence, and those are different categories.

7.4 The evidence accounting is stated, not implied

Artifact	Count	Grade
Written statements in MANIFESTO.md	73	Written proof (human-checked)
Plus the supplementary Fermat study	+14 = 87	Written proof, restricted domain
Lean 4.22.0 declarations, indi- vidually named in the Evidence Appendix	20	Formal proof
Executable verification suites (13 Python, 7 Node)	20	Finite test
Executable experimental packs	9	Model-internal finite test
Unexecuted research specifica- tions	3	Specification only
Claims in the Alpha memory fab- ric	0	—

The gap between 73 and 20 is not hidden; the Evidence Appendix names every Lean declaration and states what is **not** covered — U01, the O07 bound, O08R, the geometry, and every implementation. Most projects with a formalisation effort report the 20 and let the reader assume coverage. This one enumerates the shortfall.

The one defect in the accounting: the Manifesto appendix still says the runner requests *seventeen* suites. It requests twenty. Fix the book.

8 Steelman: the best version of the contribution

Here is the strongest honest case, made as if I were arguing for the candidate.

8.1 The contribution is a *precondition*, not a construction

Almost every framework for abstraction gives you a construction: here is the quotient, here is the minimal automaton, here is the reduced model. RPRM inverts the order. It gives you an **admission procedure that you must complete before you are allowed to propose a construction**: name the carrier, name the equality, name the question, name the ports you have and the ports you lack, name the receiver, name the operations and their enabledness, name the coverage boundary, name the hostile case, name the grade.

The mathematics that comes out the other end is ordinary. That is not a defect, it is the design. The claim is that **the ordinary mathematics is applied wrongly at a measurable rate because the preconditions are not declared**, and Section 4.6 puts a number on one instance of that: between 22% and 40% of finite partial machines admit a reduction that passes the observation-and-successor test and fails the enabledness test, and no amount of bounded observation-only validation detects it.

8.2 The four primitives that are genuinely doing work

1. **The complete fiber discipline.** NONE, ONE and MANY are reserved for closed searches; unfinished is OPEN. This is a stronger requirement than the literature's usual "no solution found." It is why the BSD HOLD is a HOLD and not a refutation.
2. **Enabledness as a first-class condition.** Standard for bisimulation specialists; systematically dropped by everyone else. The census measures the cost.
3. **Occurrences distinct from values.** Enforced at the type level, not by convention. Three admission tests in `checks/futures.py` reject value-equal, occurrence-distinct keys.
4. **Joint missing ports stay joint.** The fiber over a set of unknown ports is not the product of the marginal fibers. Most informal reasoning silently marginalises.

8.3 The process contribution, which may outlast the mathematics

A single human with no mathematical training, working almost entirely by voice with language models over roughly seven weeks, produced: a 73-statement manuscript, a 20-declaration Lean development, 20 passing verification suites, a nine-pack experimental index with complete finite censuses, a documented withdrawal, and a documented HOLD at an exact contradiction with fitted repair declared inadmissible in advance.

The failure modes that discipline is designed against — overclaiming, fitted repair, ungraded assertion, hash-as-proof, unfinished-search-as-negative-result — are precisely the failure modes of AI-assisted research generally. If the mathematics is eventually judged derivative, the operating procedure is still a reproducible answer to a live question, and I would defend publishing it on its own.

8.4 The best single sentence

RPRM is a checkable answer to the question "what does this representation keep, and how would you know if it dropped something you needed?" — and it supplies the hostile case that answers it.

9 Criticisms

Sharp, as requested. These are the things a committee would press on, and I am pressing on them rather than pretending they are not there.

9.1 The source-level yield is low relative to effort

Roughly 900 files under `research/bsd-*`, an 18.8 GB session record, 198 commits in a companion implementation repository, 982 Claude session files, 4,055 Codex rollouts. Out of that: 73 written statements, of which the great majority re-present known finite mathematics, and 20 formal declarations.

The ratio is not damning by itself — exploratory work is supposed to be wasteful — but it has a specific cause worth naming: **the abduction lane produces faster than the verification lane consumes**. The memory fabric's `claims` table has zero rows. 1,932 occurrences, 1,817 relations, **no claims**. The infrastructure for converting an occurrence into a graded claim exists and has never been used. That is the yield problem in one number.

9.2 Known-mathematics synthesis presented adjacent to novelty

The repository is careful about this in its own documents — the backlog literally writes *"Credit established coding theory"* and *"established mathematics under a project receiver reading, not a novel theorem claim."* But the careful grading lives in the backlog and the recovered-concepts notes, not in the front matter. A reader arriving at `MANIFESTO.md` sees 73 theorem-numbered statements and no column saying which are restatements.

Concrete fix, cheap: add a fourth column to the Evidence Appendix table with values in {new-as-stated, re-presentation, established-with-citation}. You already know the answers. Not publishing them is the only thing making the project look like it does not.

9.3 The metaphor umbrellas are a genuine liability

"Dark World," "prime shadows," "true pi," "six is fake," "fiving," "prestige," "the math rail and the physics rail." You know this — *"it's the most, like, numerology-coded, unfortunately"* — and you have said the operational part will be hardest to grasp for exactly this reason.

The liability is not that the terms are silly. It is that **one name currently covers several distinct contracts**, and the project's own backlog says so for Prestige (two PROMOTE contracts versus a remembered three-type taxonomy) and for Fiving (`FIVE.future` versus oriented $+5 \bmod 10$ versus lift/inverse). An umbrella term that covers three contracts will eventually be used to transport a result from one contract to another, and that is how frameworks acquire false theorems. The backlog anticipates this and forbids it. The forbidding is not enforced by any checker.

9.4 The physics hypothesis still needs independent dynamics

The two-path lab has 31 tests and a rational-component model and explicitly no physical experiment. The Manifesto III.9 states the receiver-sufficiency result correctly. The YM2 work is a finite lattice statement "through its stated order," explicitly not all-order and explicitly not universal.

None of this is overclaimed in the repository. But the *hope* attached to it is visible throughout the session record, and the hope is doing work in how the project is organised — the YM2 directories are numerous and the independent- dynamics derivation does not exist. Until a dynamics is derived rather than matched, every physics reading is an interpretation, which in this project's own grading scheme licences nothing.

9.5 The newcomer trap: one name, four things

This is the criticism I would lead with if I were a referee.

"RPRM" currently denotes, simultaneously:

1. a book (MANIFESTO.md, 73 statements),
2. a methods discipline (the contract, the handbook, the agent charters),
3. a physics hope (two-path, YM2, "what crosses the boundary"),
4. a BSD campaign (900 files, one HOLD).

A reader who samples (4) concludes the project is a crank operation grinding a millennium problem. A reader who samples (2) concludes it is a thoughtful methods paper. A reader who samples (3) concludes it is physics-adjacent speculation. A reader who samples (1) concludes it is a competent survey of finite quotient mathematics with unusual framing. **All four readers are partially right and none of them will read the other three.**

This is a naming and packaging failure, not a mathematical one, and it is the highest-leverage fixable problem in the whole corpus. Four names, four front doors, four audiences. The methods discipline in particular is being hidden behind the BSD campaign, and it is the part that is finished.

9.6 Smaller things that will be found in ten minutes

- The Manifesto appendix says seventeen suites; the runner has twenty.
- `experimental/shadow-lens/` contains a complete 181,447-scene census with 11 tests and is not in the nine-pack index that sits one directory above it.
- `experimental/process-mechanics/` is likewise unindexed.
- The fabric's `claims` table is empty, so `trace_claim` and `show_conflicts` cannot function. Either populate it or remove the interfaces from the skill.
- Lean is `NOT_RUN` in the independent verification pass. Nobody outside the project has executed the formal suite.

9.7 The criticism I decided not to make

I am not going to argue that finite examples cannot support a general framework. Every row in Section 4.6 and Section 4.8 is a complete enumeration inside a stated family, the families are stated, and the project never calls them theorems. That is the correct use of finite computation and it should be defended, not apologised for.

10 Forgotten or unfinished work worth keeping

Ranked by what I would lose sleep over if the repository were pruned tomorrow.

10.1 Shadow Lens, and it is not in the index

`experimental/shadow-lens/` contains a complete finite census: **181,447 sources for a fixed shadow, which factors as $7 \times 161^2 = 7 \times 25,921$ — seven probe fibers of exactly 25,921 each.** Eleven model tests, including 'complete fixed-shadow census: 181447 scenes, seven probe fibers of 25921'.

This is a closed MANY-fiber computation with a clean factorisation, and it is the most literal possible instance of the project's founding question: given a shadow, what is the complete set of scenes that could have cast it? It is missing from the nine-pack table in `experimental/README.md`, which sits in the parent directory.

Action: index it. It is a five-minute edit and it adds a finished exhibit to the public face of the project.

10.2 The 28-of-35 census

Three calibrated independent binary checks generate seven check functions, and **exactly 28 of the 35 possible triples preserve all eight states.** The seven failures are retained as counterexamples to "any three will do." The seven teacher-answer words form the binary simplex code, dual to Hamming-7.

The coding theory is established and the backlog says to credit it. The *census* — which 28, which 7, and why the 7 fail — is the project's own and is a complete finite result. Keep it as an exhibit.

10.3 The $-3/5$, $+2/5$ expansion/compression stack

34,936 passing assertions, exact rationals, and a non-constant $2, \ 3, \ 2$ refinement/coarsening sequence in which later coarsenings are lawful **because the stronger futures were explicitly retired.** That last clause is the interesting one: it is a worked example of lawful information destruction, which is much rarer in the literature than lawful information preservation. The open obligation is the scaling statement.

10.4 Audit105: the movable cut

A completed finite construction for moving a lens cut while preserving a relation, with $+0.4$ and -0.6 as the same phase mod 1 on different sheets of the covering line. This is the technical core of Seam Zero and it is closed. The doctrine wrapped around it is not, but the construction should not go down with the doctrine.

10.5 The unexecuted research packs

`research-packs/folding-dynamics`, `brca1`, and the immune pack. Three specifications with mathematical controls and pre-registered evaluation contracts and no data. The specifications are good — pre-registering the evaluation before touching data is the right procedure — and they cost nothing to keep. They should not be advertised, and the repository correctly does not count them among the nine packs.

10.6 The `rprm-will-lens` fitting gap

The subagent is fitted on roughly 266,000 words. The Codex corpus alone contains 1,150,000 words of your speech, and that excludes ChatGPT entirely. If the understudy is worth having, it is worth refitting on $4.3\times$ the data. That is a mechanical improvement available today.

10.7 The empty claims table

Not forgotten work exactly — infrastructure built and never used. The fabric can store claims with provenance and conflicts. Zero claims exist. Every graded statement in the Manifesto is a candidate row. Populating it would turn the fabric from a search index into the adjudication layer it was designed to be, and would make `trace_claim` and `show_conflicts` mean something.

11 Thesis-defence questions for you

Fifty-one questions. Every one is answerable in a sentence or a number; none is rhetorical. The five I would answer first are **1, 7, 18, 31 and 44**.

11.1 A. The pick and direction

1. **Chapter 12 picks the fragility spectrum over the enabledness census, the liar/teacher-Hamming paper, and the YM2 continuation. Is that the cut you would make?** If not, which of the three runners-up, and what do you see in it that I did not?
2. Is "uniform binning is the most fragile abstraction" a sentence you would put your name on if the proof completes, or does it feel too small to be worth a paper?
3. You said the proof is in the pudding and the rest will come. Would a proved extremal theorem with a counterintuitive applied corollary count as pudding, or does it have to be a millennium problem?
4. How much of your remaining Codex budget is committed to BSD, and for how long?
5. If Codex's BSD campaign never clears the 3 versus 1 HOLD, what is your stopping rule? Is there one?
6. Would you rather have one finished small result this month or a plausible large one in six?

11.2 B. The contract

7. **Do you accept that Oo5 is strong bisimulation for deterministic partial systems, and that the contribution is the precondition discipline rather than the construction?** This is the single claim in this document with the largest downstream consequences, because it determines whether the project sells novelty or sells rigour.
8. Is the preservation law intended as a biconditional in all cases, including infinite carriers, or only where the fiber is constructively enumerable?
9. Is there a fourth condition you believe Oo5 is missing, or do you regard {observation, enabledness, successor} as complete?
10. When you say "keep correlated missing ports joint," do you have a worked case where marginalising them gives a different answer? I could not find one with a computed fiber.
11. Should OPEN be allowed to carry a lower bound — for instance "at least MANY(3) but the search is unfinished" — or does that dilute the discipline?
12. Is a retraction ever acceptable where the contract asks for an inverse, or is that always a downgrade that must be stated?
13. Does "admitted context" include the time at which equality is taken, or is that a separate declaration?

11.3 C. The lenses

14. Shadow Lens computes $181,447 = 7 \times 161^2$ scenes in seven probe fibers of 25,921. Is the factor of seven structural, or an artifact of the probe choice?
15. Was Shadow Lens left out of the nine-pack index deliberately, or is that an oversight I should fix?
16. For the 28-of-35 census: do you have a characterisation of the seven failing triples, or only the list?

17. Seam Zero: do you still believe "negative five is the math rail; six and four are the physics rail," given that the phrase has not been located in source?
18. **Dark World and prime shadows appear only in chat logs and the August master context guide, with no module, checker or contract. Is there a typed version I missed, or is this abduction-lane dialect?** If it is dialect, may I mark it as such in the repository so newcomers stop treating it as a component?
19. Tile / Board / Atlas: name one complete worked instance, including a failed landing. The backlog asks for exactly this and nothing satisfies it yet.
20. Prestige: are the two PROMOTE contracts genuinely two, or two of a remembered three? If three, what is the third?
21. Fiving: give me the terminology-to-contract map in one table and the umbrella closes permanently.
22. Is the "bubble" function $(1-x^2)(1-y^2)$ the same object as the recovered "vacancy adapter"? Section 4.8 shows it is exactly what makes eight ring probes insufficient, which is the cleanest teaching case in the corpus.
23. Do you agree the ternary-grid and graded-cube results should be used as exhibits rather than pursued as research, given your own Alpha grading?

11.4 D. Evidence and formalisation

24. The Manifesto appendix says seventeen suites; the runner has twenty. Do you want that fixed in the book, or noted as an erratum?
25. Has anyone outside the project executed the Lean suite? The independent lane marked it NOT_RUN.
26. Which of the 73 written statements would you most want formalised next, if exactly one more Lean declaration were possible?
27. Would you accept a fourth column in the Evidence Appendix marking each statement as new-as-stated, re-presentation, or established-with-citation?
28. Is there any statement among the 73 that you now believe is wrong?
29. The 14 Fermat statements: do they stand as restricted results, or should they be withdrawn with the overclaim?
30. What would you accept as independent verification — a referee, a second implementation, a formal proof, or an external application that works?

11.5 E. Process

31. **The fabric has 1,932 occurrences, 1,817 relations and zero claims. Was the claims layer never used, or used elsewhere and not seeded here?** The abduction-to-verification ratio is the yield problem, and this is its measurement.
32. Would you refit `rprm-will-lens` on the full 1.15 M-word Codex corpus rather than 266 k, or do you prefer it under-fitted so it stays a pointing?
33. Is the druid/wizard/shadow-will trio working as you intended, or has the wizard lane been starved?
34. How do you decide when abduction stops and typing begins? Is there a signal, or is it budget?
35. Do you want agents to ask before spawning subworkers? `preferences.md` says no; several sessions ask anyway.

36. What is the actual status of the ChatGPT corpus — can you export it, and do you want it ingested?
37. Is OneDrive still unlinked? ChatGPT\RPRM-Spark was written on 16 September at 17:08, on the tree you said you disabled.

11.6 F. Packaging

38. Would you split the four things currently called RPRM — book, methods discipline, physics hope, BSD campaign — into four names with four front doors?
39. If exactly one of those four is the public face, which?
40. Is the methods discipline publishable on its own, as an AI-assisted research protocol paper, independent of whether the mathematics is novel?
41. Who is the intended first reader: a mathematician, a physicist, a systems modeller, or a methodologist? The repository currently addresses all four.
42. Is "RPRM" a name you want to keep?

11.7 G. The pick, in detail

43. The fragility closed form counts partial unary maps that a partition survives. Is arity 1 the right primitive for your intuition, or should the headline be stated at arity 2?
44. **"Done" for the pick is: a written proof of the extremal theorem, a complete exhaustive table, one hostile case, and a literature-priority answer. Is that the right definition of done, or do you want the applied refutation attached before you would call it finished?**
45. Would you rather I prove the extremal theorem for all (n, m) , or push the exhaustive search to $n = 40$ and publish the conjecture with the data?
46. The applied corollary says uniform binning is the worst m -block choice under unknown dynamics. Do you know a field where that would land hardest? My guess is reaction-network coarse-graining; yours will be better.
47. Do you want the census and the fragility spectrum in one paper or two?

11.8 H. Direction

48. What would make you stop working on RPRM?
49. What would you want to be true about it in five years that is not a proof?
50. Is there a result you have already seen in your head that nobody has typed yet? Chapter 10 lists what I could find; I assume it is incomplete.
51. What did I get most wrong in this document?

12 The pick

One target, not BSD, argued. I own this choice and I will be wrong in public if it is wrong.

12.1 The pick in one paragraph

Make RPRM's coverage boundary into a computable number: the fragility spectrum of a quotient under unknown operations. RPRM insists that an operational quotient is only valid relative to a *declared* operation set. Every practising modeller knows their operation set is incomplete, and the framework states that as a caution rather than as a quantity. So define, for a partition c of a finite carrier, the count $\sigma_a(c)$ of arity- a partial operations that c survives in the exact Oo5 sense, and the fragility $F_a(c) = 1 - \sigma_a(c)/(\text{ambient})$. I derived a closed form, verified it by exhaustive brute force, and found an extremal law with zero counterexamples up to $n = 24$ (*Appendix F: later the same night this became a theorem, at every arity*): **among all ways to compress n states into m blocks, the most robust is one big blob plus singletons, and the least robust is uniform binning** — the thing everyone actually does. At $n = 20$, $m = 10$ the blob survives an unknown operation 1.7×10^7 times more often than the balanced partition. That is a new invariant, a provable extremal theorem, a counterintuitive applied corollary, and a finite exact computation, all inside RPRM's own contract and none of it BSD.

12.2 The typed claim

Carrier. A finite set s , $|s| = n$, elements distinguishable, equality exact, occurrences distinct from values.

Object. A partition c of s into m blocks of sizes $(n_1, \dots, n_m)$.

Operation. A deterministic **partial** map $g : S^a \rightarrow s$ of arity a . The ambient space has $(n+1)^{(n^a)}$ elements: each of the n^a argument tuples maps to one of n targets or is undefined.

Survival (the Oo5 test, restricted to one new operation). c survives g when for all argument tuples that are componentwise c -equivalent,

1. both are in $\text{dom}(g)$ or neither is — **enabledness agreement**;
2. their images lie in the same block — **successor agreement**.

Observation agreement is inherited from c and is not re-tested.

Requested readout. $\sigma_a(c)$, the exact integer count of surviving operations, and $F_a(c) = 1 - \sigma_a(c)/(n+1)^{(n^a)}$ as an exact rational.

Supplied ports. Block sizes only. **Missing ports.** Everything else about the system — its existing alphabet, its observations, its transition structure.

Missing ports that must stay joint. The block sizes are jointly constrained by $\sum n_i = n$. Marginalising them destroys the invariant entirely; the whole content of the extremal theorem is about how the mass is distributed.

12.3 The results, computed for this document

*Evidence grade: closed form is a derived synthesis with a written proof sketch, confirmed by exhaustive finite test; the extremal law is a **conjecture** with exhaustive finite evidence to $n = 24$. Recomputed here, `_tools/fragility.py`, integers and Fraction only.*

12.3.1 Result 1 — the closed form

```
sigma_1(C) = prod_{i=1..m} ( 1 + sum_{j=1..m} n_j ^ n_i )

sigma_a(C) = prod over all a-tuples (i_1..i_a) of
  ( 1 + sum_{l=1..m} n_l ^ (n_{i_1} * ... * n_{i_a}) )
```

Why it is true, in one paragraph. Fix a block B_i (or, at arity a , a block of argument tuples $B_{\{i_1\}} \times \dots \times B_{\{i_a\}}$, of size $n_{\{i_1\}} \dots n_{\{i_a\}}$). Enabledness agreement forces g to be either undefined on the whole block or defined on the whole block. Successor agreement forces the image of the whole block to lie inside a single block B_1 , and within that constraint g may be any function from the block to B_1 . So the choices for that block are $1 + \sum_l n_l^{|block|}$, and blocks are independent. Multiply.

Null declared before the run. The discrete partition (all singletons) must give $\sigma_1 = (n+1)^n$ exactly, i.e. $F = \emptyset$: a partition that distinguishes everything cannot be broken by anything. Verified for $n = 1..8$: 2, 9, 64, 625, 7776, 117649, 2097152, 43046721 — equal to $(n+1)^n$ in every case. The instrument is calibrated before it is used.

Exhaustive confirmation against brute force, testing every single partial map:

n	set partitions	arity-1 maps tested per partition	closed form vs brute force
1	1	2	all agree
2	2	9	all agree
3	5	64	all agree
4	15	625	all agree
5	52	7,776	all agree

n	set partitions	arity-2 tables tested per partition	closed form vs brute force
1	1	2	all agree
2	2	81	all agree
3	5	262,144	all agree

12.3.2 Result 2 — the spectrum is not monotone in compression

The full spectrum at $n = 6$, every partition shape, exact rationals over $7^6 = 117,649$:

profile	blocks	sigma_1	F (exact)	F
(6)	1	46,657	70992/117649	0.603422
(5,1)	2	21,889	13680/16807	0.813947
(4,2)	2	5,733	2284/2401	0.951270
(3,3)	2	3,025	114624/117649	0.974288
(4,1,1)	3	12,691	306/343	0.892128
(3,2,1)	3	3,885	16252/16807	0.966978
(2,2,2)	3	2,197	115452/117649	0.981326
(3,1,1,1)	4	10,633	312/343	0.909621
(2,2,1,1)	4	5,929	2280/2401	0.949604
(2,1,1,1,1)	5	21,609	40/49	0.816327
(1,1,1,1,1,1)	6	117,649	0	0.000000

Read the first row against the seventh. **Total collapse to a single block is less fragile (0.603) than the balanced three-block partition (0.981).** Fragility is *not* monotone in how much you compress. The most dangerous abstractions are the moderately aggressive balanced ones, not the extreme ones. I did not expect that and it is the kind of shape that makes a referee read the next page.

12.3.3 Result 3 — the extremal law (conjecture, exhaustive to $n = 24$)

Declared before the search: for fixed n and block count m , the maximum of σ_1 is at the maximally unequal profile $(n-m+1, 1, \dots, 1)$ and the minimum is at the balanced profile (block sizes differing by at most 1).

Result: zero counterexamples. Every (n, m) with $2 \leq n \leq 24$ and every profile enumerated exhaustively. A sample:

n	m	profiles	argmax observed	predicted	argmin observed	predicted
4	2	2	(3,1)	OK	(2,2)	OK
6	3	3	(4,1,1)	OK	(2,2,2)	OK
8	4	5	(5,1,1,1)	OK	(2,2,2,2)	OK
9	3	7	(7,1,1)	OK	(3,3,3)	OK
9	5	5	(5,1,1,1,1)	OK	(2,2,2,2,1)	OK

Hostile case, run deliberately to try to break it: does the law survive at arity 2, where the block-of-tuples sizes are products rather than sums? Tested exhaustively for $2 \leq n \leq 12$, every m , comparing $\arg\max/\arg\min$ under σ_1 against σ_2 . **Zero disagreements.** The extremal shape is the same at both arities. That is stronger evidence than I expected and it is also the clue for the proof: the argument should go through for any arity because it depends on the convexity of $k \rightarrow n_1^k$, not on the arity.

12.3.4 Result 4 — the applied corollary, which is the sentence that travels

A modeller compresses n states into m blocks and does not know the full operation set. The probability that an unknown partial unary operation preserves the abstraction:

n	m	balanced binning	one big blob	blob / balanced
6	2	0.0257	0.1861	7.2
6	3	0.0187	0.1079	5.8
8	2	0.00611	0.1722	28
8	4	0.00194	0.05299	27
10	2	0.001507	0.1643	109
10	5	0.0001575	0.02634	167
12	6	1.048e-05	0.01312	1,252
16	8	2.890e-08	0.003267	113,000
20	10	4.825e-11	0.0008145	1.7×10^7

Uniform binning is the most fragile m -block abstraction under unknown dynamics, and the penalty grows superexponentially in the carrier size. Uniform binning is what almost everyone does by default: equal-width histograms, equal- population strata, balanced clusters, k -means with balanced initialisation, equal-size mesh cells. This result says that default is the worst available choice by exactly the criterion RPRM cares about.

12.4 Why this pick and not the others

12.4.1 Runner-up 1 — the enabledness census. *REJECTED as the headline, KEPT as calibration.*

The census in Section 4.6 is good work and the numbers are real: 22-40% of finite partial machines admit an enabledness-blind fold, max exposure depth exactly $n-1$, null control clean on 29,675 total machines. I ran it first and I nearly picked it.

Why it loses: the phenomenon is classical. The difference between self-loop completion and error-state completion of a partial automaton, and the $n-1$ Moore bound on distinguishing words, are both in the literature from the 1950s and 60s. A referee would call the census a well-executed exercise. It is the right *calibration* for the fragility work — it shows the enabledness condition has real bite before I start counting how it breaks — and Chapter 4 uses it that way. It is not the flagship.

12.4.2 Runner-up 2 — liar/teacher + Hamming, the 28-of-35 paper. *REJECTED as research, ENDORSED as the next paper.*

Your own backlog puts this at the top, with THEORY P1-P10, a complete three-bit census, retained counterexamples, and the simplex/Hamming-7 duality.

Why it loses as a research target: the backlog itself says "*Established coding theory, newly connected here to the recovered model*" and "*Credit established coding theory.*" You already graded it. The contribution is the connection and the recovery, which is editorial work, not new mathematics. That is a genuinely good paper and you should write it — but you asked for the target most likely to produce a legitimate new proof, and a paper whose own backlog entry says "credit established coding theory" is not it.

12.4.3 Runner-up 3 — YM2 finite lattice covering. *REJECTED, and I want to be explicit about why.*

The brief allowed a millennium-adjacent target if the finite work genuinely supports a scoped claim. It does not, and the repository says so: the result is a connected cancellation **through its stated order**, explicitly "*not an all-order or universal compression theorem.*"

Why it loses: the finite lattice statement is proved and parked. The only newsworthy step is the continuum limit, and that needs a new lens, external referees, and a level of analytic machinery that is not in the corpus and is not in your toolkit. Picking it would mean spending months to arrive at "we still need a new idea." That is exactly the shape of the BSD campaign, and you asked for something that is *not* that shape. I am rejecting it for the same reason you rejected re-auditing the toric 3 versus 1.

12.4.4 Runner-up 4 — the ternary grid / graded scar cube. **REJECTED on your own authority.**

Both ingest lanes recommended it. I recomputed every number and they are exactly right. Then the memory fabric returned your own August grading: "*established finite-dimensional mathematics, not a novelty or physics claim*" and "*established mathematics under a project receiver reading, not a novel theorem claim.*"

You were right in August and the ingest lanes were wrong last night. Correct and known is the worst combination for a flagship. It stays as the best teaching exhibit in the corpus — Section 4.8's bubble row is the finest single illustration of the aperture contract anywhere in the project — and it is not a research target.

12.5 What is actually new here, stated at the exact ceiling

I will not oversell this. Precisely:

Component	Status
The closed form for $\sigma_a(C)$	Elementary. Derivable in five lines once the question is asked. New as stated in this corpus ; priority against the wider literature is OPEN and must be checked.
The extremal law (unequal maximises, balanced minimises, fixed m)	Conjecture with exhaustive evidence to $n = 24$ and hostile-case survival at arity 2. Not proved. A proof is the first deliverable.
Non-monotonicity of fragility in block count	Finite observation , exact, surprising, not yet explained.
The applied corollary about uniform binning	Derived synthesis . Its force depends on the uniform-prior-over-operations model, which is a modelling choice that must be stated and defended.
Connection to RPRM's coverage boundary	Interpretation , and the reason the question got asked at all.

The honest summary: **this is not a deep theorem. It is a well-posed new question with a clean answer, a surprising shape, and an applied consequence that contradicts standard practice.** That combination gets read. A deep theorem in a field you do not have entry to does not.

12.6 The hostile cases, stated in advance

1. **Arity.** Does the extremal law survive at arity 2? *Already run: yes, $n \leq 12$, zero disagreements.* At arity 3?
2. **Total operations.** Restrict to total maps (no undefined). The count becomes $\prod_i \sum_j n_j^{\{n_i\}}$. Does the extremal shape change? **Not yet run.** If it flips, the result is about partiality specifically, which is a *better* story for RPRM, not a worse one.

3. **Non-uniform priors.** The corollary assumes every partial operation is equally likely. A modeller with a structured operation pool has a different prior. Does the extremal shape survive a prior concentrated on injective maps? On idempotents? **Not yet run. This is the strongest objection and it should be attacked first.**
4. **Non-determinism.** RPRM's Oo5 is deterministic. The relational case needs a different survival condition and probably a different answer.
5. **The trivial reading.** A critic will say "of course singletons are robust — you defined robustness so that distinguishing everything is free." The answer is that the law is about *fixed* m , where the compression budget is held constant, and that the non-monotonicity in Result 2 shows the criterion is not degenerate.

12.7 What "done" looks like

Milestone	Disposition when complete
Closed form for $\sigma_a(C)$, all arities, written proof	ONE(formula)
Extremal law for $\arg\max$ and $\arg\min$, all (n, m) , written proof	ONE(profile) each, or a counterexample, which would be a fine result too
Exhaustive tables to the computational limit	ONE(table) per (n, m) cell; OPEN beyond
Total-operation variant	ONE(formula) + comparison
Injective / idempotent / structured priors	MANY(family) of extremal shapes, indexed by prior
Literature priority	ONE(citation) if it exists, or NONE found after a stated search, never a silent claim of novelty
The applied refutation: a published reduced model that the fragility test flags	ONE(witness) or OPEN

Anything not on that list stays **OPEN**. The search is not finished until the enumeration is complete inside a stated family, which is your rule, not mine.

12.8 Why this is the best cut *for you specifically*

- **It is numbers.** Every claim in it is an exact integer or exact rational in a complete table. You can read it the way you say you read things.
- **It is relation-first.** The invariant is a property of a partition — a relation — not of a system. That is the project's native object.
- **It has a hostile case built into the question.** The extremal law is stated so that a single counterexample kills it, and I ran two adversarial variants already.
- **It is small enough to finish.** The closed form took an evening. The extremal proof is a term's worth of work for a good combinatorialist, or a week for you and a wizard-lane agent with exact arithmetic.
- **It is not BSD, and Codex can keep grinding BSD in parallel.** No shared resource, no shared contradiction, no fitted repair anywhere near it.
- **It makes RPRM's central warning quantitative.** Right now the framework says "your quotient is only valid relative to your declared operations" and a modeller says "sure, fine." With this, the framework says "and here is the probability yours survives, and here is why your binning choice was the worst available one." That is the difference between a caution and a result.

12.9 How I could be wrong

Three ways, ranked.

The literature check kills priority. Counting partition-respecting partial maps is elementary enough that it may exist in semigroup theory, universal algebra, or the transformation-semigroup literature under a different name. If it does, the closed form is a citation and the extremal law may still be new. *This must be checked before anything is published, and "I could not find it" is NONE-after-stated-search, not novelty. (Appendix G reports the review that was owed here, and it went against the project.)*

The uniform prior is indefensible. If a referee decides that counting all partial maps equally is not a meaningful measure, the applied corollary evaporates and only the combinatorics remains. Mitigation: hostile case 3, run early.

You do not care. This is a combinatorics result with a modelling moral. If what you want is a millennium problem, this is not that, and no amount of exact rationals makes it that. I picked it because you also said you were fully open and because you said the proof is in the pudding — and this pudding is finishable.

13 Appendix A — Portable briefing: paste this into another model

Everything below is self-contained. It assumes no prior conversation. A model reading only this appendix should be able to work with William Foster on RPRM without needing to be corrected on the same points again.

13.1 A.1 Who you are working with

William Foster. No formal mathematical training past early high-school algebra; says so himself, repeatedly and without embarrassment. Has ADHD, works almost entirely by voice dictation, and produces long associative bursts that read badly in transcript and are usually pointing at something real.

He is not asking you to flatter him and he is not asking you to protect him from negative results. He built an agent whose charter is to declare the null hypothesis before looking at data. Treat him as a collaborator with an unusual skill profile: **very strong structural pattern detection over complete numeric displays; no ability to derive standard results from memory.** Supply the derivations. Supply the complete tables. He will supply the structure.

His own framing, in his words: *"I'm wisdom guy not an int guy,"* and *"intuition is a capability detector, not a truth finder."* He already knows pattern recognition is not proof. Do not explain it to him.

13.2 A.2 The ten standing corrections

1. **Show numbers, not prose about numbers.** Give the complete set, in one view, with enough framing to know what it is of. Never "roughly a third" when you could show twelve rows.
2. **Source is not truth.** Explaining how something arose does not answer why this exact one. Say which question you answered.
3. **Do not lecture him that pattern recognition is not proof.** He polices his own over-claiming and has withdrawn a claim in public.
4. **No hedging during abduction; exact claim ceilings at publication.** Know which lane you are in and commit to it. Hedging as texture is worse than being wrong.
5. **Canonical, upstream, standard, generated, 404 and third-party are not truth vetoes.** Where something came from does not settle whether the relation is real.
6. **RPRM is not a theory of everything.** He said it once and withdrew it. Do not reinstate it for him.
7. **Equal values are not equal occurrences.** Two occurrences of 5 may be different objects. His code enforces this at the type level.
8. **NONE / ONE / MANY are only for complete solved fibers. An unfinished search is OPEN.** Reporting an incomplete search as a negative result is the error he most reliably catches.
9. **Every claim carries a domain and an evidence grade.**

10. **Do not take his words as scripture.** His own instruction: *"don't take my words as a Bible ... even do the things I didn't say."* The nine rules above are failure modes already paid for, not a cage.

13.3 A.3 What RPRM is, in 200 words

RPRM is a discipline for stating and auditing what a representation keeps. Its object is the pair (question, representation). Its central law: a representation preserves a question exactly when that question is constant on each representation fiber. An *operational* representation additionally requires matching enabledness and retained successor behaviour.

Before a claim is admitted you must declare: carrier, types, equality, admitted context; supplied ports, missing ports, requested readout; operation kind, direction, enabledness; receiver (observations and continuations to preserve); inverse, retraction, or complete preimage fiber; coverage boundary, a distinguishing hostile case, and an evidence grade.

The mathematics underneath is mostly known finite mathematics — Moore quotients, partition refinement, lumpability, Möbius inversion, barycentric coordinates, second-difference kernels, simplex/Hamming duality. The contribution is the precondition discipline, not novelty of theorem. The project says this about itself in its own afterword.

It is **not** a theory of everything, not a physics result, not a proof of BSD or Fermat, and not finished.

13.4 A.4 Vocabulary you will meet

Term	What it means
Carrier	The declared set of values plus the equality used on it
Port	A coordinate of a relation; supplied or missing
Aperture	Which ports are supplied and what readout is requested
Fiber	The complete set of completions consistent with the supplied ports
Receiver	What must be preserved: observations, and for operational receivers also operations, timing, failures, futures, traces
Enabledness	Whether an operation is defined at a state. A first-class condition, not an implementation detail
Operational quotient (Oo5)	A representation with matching observation, enabledness and successor behaviour
NONE / ONE / MANY	Dispositions of a complete fiber
OPEN	An unfinished search. Not a negative result
Hostile case	The nearest thing that breaks the claim, stated by the claimant
Tile / Board / Atlas	An optional hierarchy of local chart, assembled charts, and global cover. Currently an umbrella with no complete worked instance
Dark World, prime shadows, fiving, prestige, true pi, Seam Zero rails	Abduction-lane dialect. Some have real constructions underneath; none is a typed module. Ask before treating one as a component

13.5 A.5 Evidence grades, in increasing force

Definition (fixes vocabulary, licences nothing) → Established mathematics (citation, not credit) → Written proof (human-checked; 73 in the manuscript) → Formal proof (machine-checked; 20 Lean declarations) → Finite test (complete enumeration inside a stated family; not a theorem) → Derived synthesis (inherits the weakest input grade) → Conjecture / OPEN → Interpretation (no mathematical force) → Process evidence (provenance only).

A hash binds bytes. It does not prove their contents. Tool self-tests do not establish their own soundness.

13.6 A.6 The working split

He runs three agent roles and they must not be mixed:

- **Druid** — abduction only. Hypothesis stream. No testing, no hedging, no register-policing. Output is never evidence.
- **Will-lens** — his own abduction moves, fitted on his transcripts. **Run it blind:** never tell it what the formal lane concluded. Output is a pointing, never a conclusion.
- **Wizard** — typing and testing. Declare the null **before** looking, compute in exact arithmetic, report plainly. Never used during abduction.

If you are being asked to riff, riff without hedging. If you are being asked to verify, verify without decoration. Ask which if it is not obvious — once.

13.7 A.7 Practical operating notes

- He authorises unattended overnight work and does not want to be asked to approve each step. *preferences.md: "use subworkers and optional tools when they would actually help; do not ask each time."*
- He works in bursts and hits usage limits. Finish something durable per session.
- Exact arithmetic, always. `fractions.Fraction`, integers, no floats in anything that will be reported as a result.
- Complete enumerations inside stated families are the preferred evidence. Say the family.
- Declare the null before running. Put the null in the output.
- Preserve counterexamples and corrections beside surviving results. Do not clean them out.

13.8 A.8 Current state, as of 17 September 2026

Item	State
Manuscript	73 written statements (+14 restricted Fermat = 87)
Formal	20 Lean 4.22.0 declarations; coverage shortfall enumerated in the appendix
Verification	20 suites (13 Python, 7 Node), all passing. <i>The book still says 17 — that is a known erratum.</i>
Experimental	9 indexed packs; shadow-lens (181,447-scene census) and process-mechanics exist but are not indexed
BSD campaign	On HOLD at an exact contradiction: normalized toric expression $\equiv 3 \bmod 5$ versus retained $U_5 = 1$. No fitted scalar repair admitted. Do not attempt a repair.
Fermat	Unrestricted claim withdrawn . 14 restricted statements survive.
Memory fabric	Populated: 115 blobs, 1,932 occurrences, 1,817 relations, 0 claims
ChatGPT corpus	No platform-native export exists. ~205,000 words survive, roughly 15% of the lane : a complete 115,299-word two-sided thread (42 user messages) plus a fidelity-labelled August export package plus smaller pasted files. A 279,537-word "CHATLOG" that an earlier pass counted here is a Codex log, not ChatGPT . 166 further handoff zips in Downloads\ are uningested.

13.9 A.9 If you are asked to pick research

Do not pick BSD. Another lane is already committed to it and the HOLD stands.

Do not pick the ternary-grid kernels or the graded Boolean-cube scar. Both are exactly correct and both are graded *established finite-dimensional mathematics, not a novelty* by the project's own August record. They are excellent teaching exhibits.

Do not pick the Yang–Mills continuum limit. The finite lattice covering is proved *through its stated order only* and explicitly is not an all-order theorem.

The liar/teacher + Hamming work (28 of 35 triples preserve all eight states; seven teacher words form the binary simplex code dual to Hamming-7) is the right **next paper** and the wrong **next research target** — the backlog itself says to credit established coding theory.

The current recommended target is the **fragility spectrum**: count the partial operations a quotient survives, prove the extremal law, check the priority literature. See Appendix B.

14 Appendix B — The pick, as a standalone specification

Reproduce this appendix verbatim into any model that is going to work on it.

Question. RPRM says an operational quotient is valid only relative to a declared operation set. Real operation sets are incomplete. How robust is a given quotient to the operations you did not declare?

Setup. Finite carrier S , $|S| = n$, exact equality. A partition c with blocks of sizes $(n_1, \dots, n_m)$, $\sum n_i = n$. A candidate new operation is a deterministic **partial** map $g : S^a \rightarrow S$; the ambient space has $(n+1)^{(n^a)}$ elements. c **survives** g iff for all componentwise c -equivalent argument tuples, (i) both or neither are in $\text{dom } g$, and (ii) the images lie in the same block.

Definitions. $\text{sigma}_a(C) = \text{number of surviving operations}$. $F_a(C) = 1 - \text{sigma}_a(C) / (n+1)^{(n^a)} = \text{fragility}$, an exact rational.

Established here (closed form; exhaustively confirmed).

$$\begin{aligned} \text{sigma}_1(C) &= \prod_{i=1..m} (1 + \sum_{j=1..m} n_j \wedge n_i) \\ \text{sigma}_a(C) &= \prod_{\text{over } a\text{-tuples } (i_1..i_a) \text{ of}} (1 + \sum_1 n_1 \wedge (n_{\{i_1\}} * \dots * n_{\{i_a\}})) \end{aligned}$$

Proof sketch: enabledness agreement forces g to be all-or-nothing on each block of argument tuples; successor agreement forces the image of that block into a single block b_1 , with any function allowed inside; blocks are independent; multiply. Confirmed by brute force over every partial map for $n \leq 5$ at arity 1 and $n \leq 3$ at arity 2.

Null control (declared before running, and it held). The discrete partition must give $\text{sigma}_1 = (n+1)^n$, i.e. $F = 0$. Verified $n = 1..8$.

Open conjecture (exhaustive evidence, $n \leq 24$, zero counterexamples). For fixed n and block count m : $\text{argmax } \text{sigma}_1 = (n-m+1, 1, \dots, 1)$ and $\text{argmin } \text{sigma}_1 = \text{the balanced profile}$. The same extremal shapes hold at arity 2 for $n \leq 12$.

Observed and unexplained. Fragility is **not monotone** in block count. At $n = 6$: total collapse (6) has $F = 0.603$, while balanced (2,2,2) has $F = 0.981$.

Applied corollary (derived synthesis; depends on a uniform prior over partial operations). Uniform binning is the most fragile m -block abstraction under unknown dynamics. At $n = 20$, $m = 10$ the one-big-blob profile survives an unknown partial operation 1.7×10^7 times more often than the balanced profile.

Hostile cases, in priority order.

1. Non-uniform priors — injective maps only, idempotents only, structured operation pools.
Strongest objection. Attack first.
2. Total operations only (drop partiality): does the extremal shape change?
3. Arity 3 and above.
4. Non-deterministic / relational operations.

5. The degeneracy objection: "singletons are trivially robust." Answer with fixed- m framing and with the non-monotonicity result.

Priority obligation. Search semigroup theory, universal algebra and transformation-semigroup literature for prior art on counting partition-respecting partial maps. Report **ONE(citation)** or **NONE after a stated search**. Never a silent novelty claim.

Done conditions. ONE(formula) for σ_a ; ONE(profile) for each extremal, or a counterexample; ONE(table) per computed (n, m) cell and OPEN beyond; ONE(citation) or NONE-after-search on priority. Everything else OPEN.

Forbidden. BSD. Repairing the toric 3 versus 1. Floating-point arithmetic in any reported number. Claiming novelty before the literature search. Upgrading the conjecture to a theorem on the strength of $n \leq 24$.

15 Appendix C — Reproduction

Three scripts were written for this document. All use integers and `fractions.Fraction` only; no floating-point arithmetic appears in any reported value except the explicitly-labelled decimal approximations in the last column of tables.

Script	Produces	Runtime
<code>_tools/center_check.py</code>	Section 4.8, 4.9: ternary-grid ranks and nullities, withheld-centre determination, graded cube kernels, the $\text{weight} \leq 2$ restriction	~4 s
<code>_tools/census.py</code>	Section 4.6: the enabledness census, exposure depths, the minimal witness, the null control	~10 s
<code>_tools/fragility.py</code>	Chapter 12: the closed form, brute-force confirmation, the $n = 6$ spectrum, the extremal search to $n = 24$, the arity-2 hostile case, the applied table	~9 s
<code>tools/fragility_hostile.py</code>	Appendix E: all five hostile cases, restricted operation classes, fixed-domain slices, arity 3, the extended search to $n = 45$	~96 s
<code>tools/fragility_priority.py</code>	Appendix E: the prior-art separation, the enabledness price, the two-monoid extremal comparison, the subset-sum characterisation	~7 s

Raw outputs are retained as `center_out.txt`, `census_out.txt` and `fragility_out.txt` beside the scripts. Every number in this document marked *recomputed* came from one of those three runs. Every number marked *reported* is quoted from a named file in `C:\github\RPRM-open` and was not re-executed here. The Lean suite was not run. The BSD interior was not read. The ChatGPT corpus does not exist locally.

16 Appendix E — Overnight addendum: the hostile cases were run, and two of them bit

Written after Chapters 1–13 were finished and the PDF was first rendered. The earlier chapters are left exactly as they were. This appendix records what happened when the hostile cases declared in Section 12.6 were actually executed, including the place where my own headline claim did not survive. Full record in `research/rprm-catchup-2026-09-17/FRAGILITY.md`; scripts in `tools/`.

16.1 E.1 What I said I would attack, and what happened

Section 12.6 listed five hostile cases in priority order and marked three of them "not yet run." All five are now run.

Hostile case	Outcome
H1 non-uniform priors — declared strongest objection	Bit. The argmin half of the law is prior-dependent
H2 total operations only	Survived. $n \leq 18$, every m , zero failures
H3 arity 3	Survived. $n \leq 10$, zero failures
H4 push the exhaustive search	Survived and strengthened: 903 cells , $n \leq 45$, zero counterexamples
H5 the degeneracy objection	Answered structurally, and Result 2's non-monotonicity holds at every $n = 4 \dots 12$

16.2 E.2 Where the claim broke

The conjecture has two halves. They do not have the same status.

Operation class	argmax = maximally unequal	argmin = balanced
All partial maps, $n \leq 45$	0 failures	0 failures
Total maps only, $n \leq 18$	0	0
Arity 2, $n \leq 12$ / arity 3, $n \leq 10$	0	0
Idempotent total maps, $n \leq 7$	0	0
Injective partial maps , $n \leq 7$	0	2 failures
Permutations , $n \leq 7$	1 failure	4 failures

The maximum side is nearly prior-free. The minimum side is not. And the minimum side is the one carrying Chapter 12's applied corollary. So:

Correction to Section 12.3, Result 4. "Uniform binning is the most fragile m -block abstraction under unknown dynamics" holds when the unknown operations may be **non-injective or partial** — that is, when the dynamics can merge states or stall. It **reverses** when the unknown operations are bijective.

The mechanism is clean enough to state in one line, and it makes the corrected claim more useful than the original. Under permutations, a partition survives exactly when the map permutes blocks bijectively, and **blocks of equal size can be interchanged**. At $n = 4$, the balanced profile $(2, 2)$ admits $2 \times 2! \times 2! = 8$ surviving permutations while $(3, 1)$ admits only $3! \times 1! = 6$, because unequal

blocks cannot swap. Balanced binning buys symmetry. Under general partial maps the opposite force dominates, because a large blob absorbs arbitrary images. Which wins is a property of the prior, and now you can tell which regime you are in.

I also ran the sharpest version of the objection — restrict to partial maps with **exactly** d defined points, which removes the "mostly-undefined maps dominate the count" explanation entirely. At $n = 6$:

d	class size	(5,1)	(4,2)	(3,3)	(4,1,1)	(3,2,1)	(2,2,2)
0	1	1	1	1	1	1	1
1	36	6	0	0	12	6	0
2	540	0	20	0	36	14	36
3	4,320	0	0	108	0	120	0
4	19,440	0	272	0	258	216	432
5	46,656	3,126	0	0	3,096	504	0
6	46,656	18,756	5,440	2,916	9,288	3,024	1,728

At $d = 6$ the law holds in both the $m = 2$ and $m = 3$ groups. At $d = 4$ it **inverts** for $m = 3$: (2,2,2) beats (4,1,1). The law is a statement about the aggregate over domain sizes and not about every slice, and saying it without that qualifier would be an overclaim.

16.3 E.3 Why the zeros are there — a new exact characterisation

Those zeros are not sampling noise. Because a block is wholly enabled or wholly disabled, **the domain size of any surviving partial operation is a subset sum of the profile**. Under the successor-only condition every domain size $0 \dots n$ is reachable.

profile	admissible domain sizes under Oo5	count	of $n+1$
(6)	0, 6	2	7
(3,3)	0, 3, 6	3	7
(5,1)	0, 1, 5, 6	4	7
(4,2)	0, 2, 4, 6	4	7
(2,2,2)	0, 2, 4, 6	4	7
(4,1,1)	0, 1, 2, 4, 5, 6	6	7
(3,2,1)	0, 1, 2, 3, 4, 5, 6	7	7
(1,1,1,1,1)	0, 1, 2, 3, 4, 5, 6	7	7

Evidence grade: written proof, immediate from the enabledness condition. It explains every zero above and it is why a blob profile concentrates its surviving operations at a few very large domain sizes where the counts are enormous.

16.4 E.4 The priority search — and the result it produced instead

Section 12.7 said the literature check might kill priority and had to run before anything was published. It ran, and the outcome is better than either possible answer I anticipated.

Prior art exists, and it is for a different monoid. The monoid of *total* transformations preserving a partition was introduced by H. Pei and enumerated for arbitrary finite partitions by Sarkar and Singh, arXiv:2006.04242. The *partial* analogue for a *uniform* partition is Cicalò, Fernandes and Schneider, arXiv:1210.4775, with published order $(m(n+1)^n - m + 1)^m$ for m blocks of size n .

But the literature condition is successor agreement only. Wherever the map happens to be defined on a block, the images must stay in one block. There is no condition on the domain. So the two counts are genuinely different objects:

$\text{sigma_blind}(C) = \text{prod}_i (1 + \sum_j [(n_j+1)^{\{n_i\}} - 1])$	literature
$\text{sigma}_E(C) = \text{prod}_i (1 + \sum_j n_j^{\{n_i\}})$	RPRM 005

Deriving `sigma_blind` from scratch and collapsing it onto a uniform profile reproduces the published order **exactly** in all 22 cases computed, including $m = 5$, $n = 3$ where both give 3,150,905,752,576. Both formulas also match exhaustive brute force over every partial map for $n \leq 6$. So the identification is verified, not assumed.

The enabledness price. $\text{rho}(C) = \text{sigma}_E / \text{sigma_blind}$ is the exact fraction of literature-admissible reductions that RPRM's condition keeps:

n	profile	sigma_E	sigma_blind	rejected by enabledness
6	(6)	46,657	117,649	60.3%
6	(3,3)	3,025	16,129	81.2%
6	(2,2,2)	2,197	15,625	85.9%
8	(4,4)	263,169	1,560,001	83.1%
8	(3,3,2)	91,287	912,951	90.0%
8	(2,2,2,2)	83,521	1,185,921	93.0%
any	discrete	$(n+1)^n$	$(n+1)^n$	0.0%

For a balanced four-block reduction of an eight-state system, **93% of the abstractions the classical monoid admits are rejected by enabledness.** The discrete-partition row is the declared null and it is exactly zero.

16.5 E.5 The flagship, which I did not see coming

Running both monoids side by side over the same 231 (n , m) cells with $3 \leq n \leq 24$:

Monoid	argmax law	argmin law
<code>sigma_E</code> — enabledness enforced	0 failures (903 cells, to $n = 45$)	0 failures
<code>sigma_blind</code> — successor only	14 failures in 231 cells	41 failures

And where they disagree they **invert**. First disagreement, $n = 11$, $m = 9$:

<code>sigma_E</code>	argmax (3,1,1,1,1,1,1,1,1)	argmin (2,2,1,1,1,1,1,1,1)
<code>sigma_blind</code>	argmax (2,2,1,1,1,1,1,1,1)	argmin (3,1,1,1,1,1,1,1,1)

The most robust profile under one condition is the least robust under the other.

On the finite family $3 \leq n \leq 24$, the classical partition-preserving partial transformation monoid has no clean extremal shape — 55 exceptions in 231 cells — while the submonoid cut out by adding RPRM's enabledness condition has an exact one, with zero exceptions in 903 cells up to $n = 45$.

Evidence grade: finite test, complete enumeration inside the stated ranges. Not a theorem. Both general statements are OPEN.

That is the strongest thing to come out of the night, and it is a better argument for RPRM than anything in Chapter 8. It does not say the enabledness condition is *correct* — that remains a modelling question. It says the condition picks out a structurally better-behaved object than the classical one, which is a reason to take it seriously that does not require agreeing with the framework's philosophy first.

16.6 E.6 Revised status of the pick

Statement	Was	Now
Closed form $\sigma_a(C)$	derived, confirmed	unchanged: ONE(formula) , written proof + complete finite test
Priority of the closed form	OPEN	σ_{blind} is ONE(citation) . σ_E was recorded here as NONE-after-a-single-pass-search. Appendix G overturns this : the review was done, and the count is a corollary of Sarkar–Singh Theorem 6.1
Extremal law	conjecture to $n = 24$	conjecture to $n = 45$, 903 cells, and shown to be specific to the enabledness-enforced monoid
"Uniform binning is worst"	derived synthesis	restricted : holds under partial, total and idempotent priors; fails under bijective priors
Subset-sum domain characterisation	not stated	new, ONE(characterisation) , written proof
Enabledness price ρ	not stated	new, ONE(rational) per profile
Enabledness restores an extremal law the classical monoid lacks	not stated	new, ONE(comparison) on the stated range; general case OPEN

16.7 E.7 What I got wrong, plainly

Chapter 12 sold the applied corollary without the prior qualifier. I declared the prior sensitivity as the strongest objection and then stated the corollary as if it had already survived it. It had not been run yet. When it was run, it failed on the bijective classes. The corollary is still true and still useful — it is now true *with a stated regime*, which is what it should have said in the first place.

Chapter 12 also treated the priority question as a threat to the result. It was the opposite. Finding the prior art is what made the comparison in E.5 possible, because you cannot say "enabledness restores a law" until you know what the law-less version is and can cite whose it is.

16.8 E.8 What is owed

1. A real literature review for σ_E . One search pass is not a review, and until it is done nothing here is novel.
2. The exchange lemma — see E.9, which was written after this list and changes the answer.
3. A characterisation of σ_{blind} 's 55 exceptions. They cluster at large m with mostly-singleton profiles, which suggests a clean description exists.
4. Take one published finite reduced model of a partial system and compute its ρ . That is the step that turns an instrument into a finding.

16.9 E.9 One open problem instead of two

Item 2 above was written as a suggestion. It turned out to be the right move and it was worth doing immediately, so the rest of the night went into it.

The reduction. The extremal law is awkward because it quantifies over a whole poset of profiles. It follows from one local statement:

Exchange lemma (conjecture). If C has blocks of sizes $a \geq b$ with $b \geq 2$, and C' replaces them by $a+1$ and $b-1$, then $\sigma_E(C') > \sigma_E(C)$ strictly.

The exchange order on profiles with fixed (n, m) is the majorisation order. Its unique maximum is $(n-m+1, 1, \dots, 1)$ and its unique minimum is the balanced profile. So the exchange lemma implies both halves of the extremal law at once, and it only ever mentions two blocks.

Half of it is now proved, not observed. Writing $s_k = \sum_j n_j^k$, the exchanged profile majorises the original and $x \rightarrow x^k$ is convex, so Karamata gives $s'_k \geq s_k$ for all $k \geq 1$, strictly for $k \geq 2$. Every factor of σ_E indexed by an unchanged block therefore weakly increases, and the grown block's factor strictly increases. *Evidence grade: written proof.* What is left is one shrinking factor, $(1 + s'_{b-1})$, and the crude sandwich bound is lossy there by exactly $(1+m)^2$. That gap is the entire remaining problem.

Evidence, which is far heavier than for the extremal law itself. Every valid exchange in every profile, $n = 4 \dots 40$:

Quantity	Value
Exchanges tested	5,686,463
Non-increases	0
Tightest ratio	1.000678, at $n = 40, (26, 7, 7) \rightarrow (26, 8, 6)$
$m = 2$ pushed to $n = 400$	39,601 exchanges, 0 failures

Where it is tightest, which is where a proof has to work. The minimising exchange always has the same shape — a large spectator block plus two *equal* blocks being split apart — and the ratio falls toward 1:

n	tightest exchange	ratio
6	$(2, 2, 2) \rightarrow (3, 2, 1)$	1.768320
12	$(6, 3, 3) \rightarrow (6, 4, 2)$	1.286139
18	$(10, 4, 4) \rightarrow (10, 5, 3)$	1.079199
24	$(14, 5, 5) \rightarrow (14, 6, 4)$	1.023515
30	$(18, 6, 6) \rightarrow (18, 7, 5)$	1.006507

A ratio heading to 1 is exactly the shape of a claim that is about to fail, so I attacked the family directly rather than extrapolate: j to 200, spectator κ to **100,000**, plus three-equal-block, all-equal-block with m to 40, and no-spectator to $j = 2000$. **Zero refutations.** Along $(2j, j, j) \rightarrow (2j, j+1, j-1)$ the excess above 1 halves cleanly each step — successive quotients 1.995, 1.920, 1.868, ..., 1.9975 at $j = 40$ — converging to 2. The ratio approaches 1 and does not cross it.

And the no-spectator case has an exact answer. With $\text{sigma_E}(j, j) = (1 + 2^{-j})^2$ and $\text{sigma_E}(j+1, j-1) = (1 + (j+1)^{j+1} + (j-1)^{j+1})(1 + (j+1)^{j-1} + (j-1)^{j-1})$, using $((j-1)/(j+1))^{j\pm 1} \rightarrow e^{\pm 2}$:

$$\begin{aligned} \text{sigma_E}(j+1, j-1) / \text{sigma_E}(j, j) &\rightarrow e^2 (1 + e^{-2})^2 / 4 \\ &= (e + e^{-1})^2 / 4 \\ &= \cosh^2(1) = 2.381097845541816\dots \end{aligned}$$

Derived first, then checked in exact rational arithmetic:

j	n	ratio	limit – ratio	j · (limit – ratio)
10	20	2.173595361539517	2.075e-01	2.0750
100	200	2.357623065853014	2.347e-02	2.3475
1,000	2,000	2.378720143406846	2.378e-03	2.3777
20,000	40,000	2.380978799147820	1.190e-04	2.3809

So $\text{ratio} = \cosh^2(1)(1 - 1/j) + O(1/j^2)$, the $1/j$ coefficient being $\cosh^2(1)$ again. **The balanced two-block partition is not marginally the worst choice — it is worse by a fixed factor of about 2.381 that does not wash out as the carrier grows.** *Disposition ONE(constant); evidence grade: written derivation plus exact finite test.*

Net change to Chapter 12. Two open conjectures became one, stated locally, with the easy half proved, the hard half isolated to a named family, and several million exchanges of evidence behind it. That is the artifact I would hand a combinatorialist. It is also the honest ceiling: the lemma is not proved, and until it is, the extremal law stays a conjecture.

The closing note that stood here has moved to the end of Appendix F, so that it is still the last thing in the document.

17 Appendix F — The lemma closed

Written in the second half of the same night, after Appendix E was already in the PDF. E ended by listing the exchange lemma as the one open problem worth handing to someone else. It did not need handing to anyone. This appendix records the proof, the reason the earlier attempts failed, and the two places where the result is narrower than it will be tempting to say it is.

17.1 F.1 Why the earlier attempts failed

Every attempt in Appendix E went through a sandwich bound: control the product over the unchanged blocks, control the two changed factors, and hope the remainder is positive. That bound is lossy by a factor of $(1+m)^2$, and the numerics said the true margin in the tight family was 1.000678. There was never going to be enough room. I spent most of the night making the numerical attack more brutal — spectator blocks to 100,000 — instead of asking why the bound was lossy.

It was lossy because it was bounding something that did not need bounding.

17.2 F.2 The observation

A profile plays **two separate roles** in sigma_E and the proof only works if you refuse to conflate them. Writing $S_k(p) = \sum_j n_j^k$,

$$\begin{aligned} f_p(k) &= 1 + S_k(p) && \text{the factor function} \\ \text{sigma}_E(p) &= \prod_i f_p(n_i) \end{aligned}$$

the profile supplies the **bases** inside f_p , and separately supplies the **exponents** at which f_p gets evaluated. The exchange $(a, b) \rightarrow (a+1, b-1)$ moves both at once, which is why it looked entangled. Move them one at a time and each move is easy.

Bases. Karamata, which was already proved in Appendix E: $(a+1, b-1)$ majorises (a, b) , $x \rightarrow x^k$ is convex, so S_k weakly increases for every $k \geq 1$ and strictly for $k \geq 2$. Hence $f_q \geq f_p$ **pointwise**. The spectator product therefore increases factor by factor and needs no bound at all. That is the $(1+m)^2$ that the sandwich was throwing away.

Exponents. $f_q(k) = e^{k \cdot \theta} + \sum_j e^{k \ln q_j}$ is a sum of exponentials, hence **log-convex** in k . The constant 1 is the $e^{k \cdot \theta}$ term — it is not a nuisance, it is one of the exponentials. It is **not** required, though: deleting it leaves another sum of exponentials, equally log-convex. (Appendix G corrects an earlier sentence here which said the result is false without it. It is not.) For a convex $h = \log f_q$ and $a \geq b$, the interval $[a, a+1]$ lies weakly right of $[b-1, b]$, so $h(a+1) - h(a) \geq h(b) - h(b-1)$. The exchange moves the two evaluation points apart at fixed sum, and log-convexity says that cannot decrease their product.

Chain them:

$$\begin{aligned} \text{sigma}_E(q) &= (\prod_c f_q(c)) f_q(a+1) f_q(b-1) \\ &\geq (\prod_c f_q(c)) f_q(a) f_q(b) && \text{exponents, log-convexity} \\ &> (\prod_c f_p(c)) f_p(a) f_p(b) && \text{bases, Karamata} \\ &= \text{sigma}_E(p) \end{aligned}$$

strict because $a \geq 2$. **That is the whole proof.**

17.3 F.3 What it gives

Theorem (exchange). For a profile with blocks $a \succeq b \succeq 2$, replacing them by $a+1$ and $b-1$ strictly increases σ_E .

Corollary (the extremal law). For fixed n and block count m , σ_E is uniquely maximised at $(n-m+1, 1, \dots, 1)$ and uniquely minimised at the balanced profile.

The corollary needs no lattice theory. If a profile is not balanced, some pair has $a \succeq b+2$, so the reverse exchange applies and strictly decreases; iterate and you terminate at balanced. If a profile is not maximally unequal, some block other than the largest has size $\succeq 2$, so a forward exchange applies and strictly increases; iterate and you terminate at $(n-m+1, 1, \dots, 1)$.

Chapter 12 and Appendix E both called this a conjecture. It is a theorem. Every step was machine-checked in exact integer arithmetic across all 128,121 distinct exchanges of all 28,622 partitions of $n = 4 \dots 30$, with a null control confirming that log-convexity holds for the unexchanged profile too — it is a property of the factor function, not an artifact of the exchange. Zero failures.

17.4 F.4 Every arity, by switching to convex order

Appendix F.2 covers arity 1. At arity r the factors are indexed by **products** of block sizes, so there are m^r evaluation points and log-convexity of two points is not enough. What is needed is: for every convex ϕ ,

$$\sum_{\text{over } r\text{-tuples } T} \phi(e'_T) \succeq \sum_{\text{over } T} \phi(e_T).$$

The instinct is to prove the exponent multiset majorises the original. That is the wrong frame, because the natural decomposition of the tuples is a disjoint union and **majorisation is not additive over disjoint unions**. But majorisation was only ever being used to get the $\sum \phi$ inequality, and *that* is additive. Phrase it as convex order and the obstruction disappears:

1. Read an r -tuple as r independent coordinate draws. Condition on which coordinates land in the two changed blocks and on the spectator product p .
2. Within a cell the exponent is p times a product of independent draws, each uniform on $\{a, b\}$ before and $\{a+1, b-1\}$ after. The single draw moves up in convex order.
3. Convex order is closed under multiplication by an independent nonnegative factor and under scaling by a nonnegative constant, so each cell moves up.
4. Sum over cells.

So the exchange lemma, and the extremal law, hold at every arity. Checked against six convex test functions the proof never mentions, with two concave controls required to reverse the inequality.

17.5 F.5 The mechanism behind Chapter 12's flagship

Appendix E's best result was empirical and unexplained: the classical successor-only monoid has no clean extremal shape, 55 exceptions in 231 cells, while the enabledness-enforced one has an exact law. I could say *that* it happened, not *why*.

The proof answers it in one line. The classical factor function is

$$f_{\text{blind},p}(k) = 1 + \sum_j (n_{j+1})^k - 1 = \sum_j (n_{j+1})^k - (m - 1),$$

a sum of exponentials **minus a positive constant** whenever $m \geq 2$. Subtracting a constant does not preserve log-convexity. So the exponent step is not merely harder for the classical monoid — **its hypothesis is false there.**

factor function	instances tested	log-convexity violations	worst ratio
enabledness en- forced	25,702	0	—
successor only	25,702	7,488	0.1357

And the violations sit in exactly the regime where the extremal law fails for `sigma_blind`: many blocks, mostly singletons, where the -1 per block is largest relative to the whole factor.

Enabledness is not decoration. Dropping the domain requirement subtracts one unit per block from every factor, and that subtraction is precisely what destroys log-convexity and with it the extremal law.

This is the strongest form of the Chapter 8 argument I can make, and it is worth being clear about what it does and does not say. It does **not** say enabledness is the correct modelling choice; that stays a modelling question, and Chapter 12 still owes an applied `rho`. It says the condition RPRM adds for modelling reasons is the condition that makes the resulting count analytically well behaved, and that is a reason to take the framework seriously which does not require agreeing with its philosophy first.

17.6 F.6 Two places this is narrower than it will sound

It is a theorem about one prior. Uniform over partial maps. Appendix E's counterexamples are untouched and still stand: under bijective priors the argmin half is **false**, and within a fixed domain size it is **false**. A proof under one measure is not a proof under all measures, and the applied corollary from Chapter 12 still carries its regime — uniform binning is the most fragile choice when the unknown dynamics can merge states or stall, and not when they cannot. The total-map and idempotent-map classes remain conjectures; the proof does not reach them, because their factor function is not $1 + \sum n_j^k$.

It is not a novelty claim. Proving a statement and establishing that nobody has proved it are different acts, and only the first one happened tonight. The literature obligation from Section 12.7 is unchanged.

17.7 F.7 Revised status, once more

Statement	Appendix E said	Now
Closed form $\sigma_a(C)$	ONE(formula), written proof	unchanged
Exchange lemma	conjecture, 5.7M exchanges	ONE(theorem), written proof, all arities
Extremal law	conjecture to $n = 45$	ONE(theorem) for the uniform partial-map prior
Why σ_{blind} lacks the law	observed, unexplained	ONE(separation) : its factor function is not log-convex
Extremal law, bijective priors	refuted	unchanged — still refuted
Extremal law, total and idempotent priors	conjecture	total maps promoted to theorem in Appendix G ; idempotents still OPEN, now with a derived obstruction
Priority	NONE-after-stated-search	overturned in Appendix G — the count is a corollary of published work; the condition is Fernandes's P-stability
Applied ρ on a published model	owed	still owed

If you read one sentence of this appendix: **the conjecture at the centre of Chapter 12 is now a theorem at every arity, the proof is four lines once you stop conflating the two roles the profile plays, and the same four lines explain why the classical monoid does not have the law.**

18 Appendix G — What two adversaries did to Appendix F

Appendix F was written at midnight and claimed a theorem. Before dawn it was attacked twice: once by a real literature review, and once by a hostile audit of the proof itself. Both attacks landed. Neither killed the theorem.

This appendix is the result. It is placed last because it is the most useful thing in the document for judging whether any of the rest should be believed — you can read what was claimed, what was checked, and what was taken back.

18.1 G.1 The scoreboard

Claim as of Appendix F	Verdict	Now
Exchange lemma is a theorem, every arity	upheld	unchanged, independently re-derived
Extremal law follows	upheld	unchanged
The closed form σ_E was not located in the literature	overturned	it is a corollary of a published theorem
The enabledness condition appears to be ours	overturned	it is Fernandes's <i>P-stability</i> , 1998
The +1 is what makes log-convexity work	overturned	it is analytically irrelevant
Product-multiset majorisation fails, so use convex order	overturned	it works, and it is the shorter proof
Total maps are outside the proof	overturned	they are inside it
Idempotents are outside the proof	upheld	and now with a derived obstruction

Six overturned claims, none of them the theorem. That ratio is the honest summary of what a night of unreviewed work is worth.

18.2 G.2 The literature review, and the two things it cost

The priority position in Chapter 12 rested on **one search pass**, and said so. A real review was owed. Done, it answers four questions, and it goes against the project on two of them.

The count is not new. Sarkar and Singh, *On certain Semigroups of Transformations that preserve a partition*, arXiv:2006.04242, *Comm. Algebra* 49(1) 2021, 331–342, Theorem 6.1, count the **total** partition-preserving maps for an arbitrary finite partition. Regrouped over blocks their formula is $\prod_i \sum_j n_j^{n_i}$ — which is σ_E with the 1 + deleted from each factor. And the 1 + is a sink:

*Adjoin * as a new singleton block. The bijection "undefined $\mapsto$ *" carries the enabledness-respecting partial maps onto exactly the total maps that fix * — a block either maps into a block, or goes wholesale to *, and {*} is itself a block, which is precisely what enabledness buys. So $\sigma_E(P) = |T(X + \{*\}, P + \{\{*\}\})| / (n+1)$.*

Verified rather than accepted: exactly, on all 507 profiles with $n \leq 14$, with Sarkar–Singh's own formula first checked against brute-force enumeration of all total maps for $n \leq 6$. Worked instance: $\text{sigma_E}(3,3) = 3025 = 21175 / 7$.

The condition is not new either. The literature calls it **P-stability**, and Caneco–Fernandes–Quinteiro (arXiv:1905.11489) attribute it to Fernandes, *Semigroup Forum* **56** (1998), 418–433: " α is P-stable if $X_{\{i_x\}} \subseteq \text{Dom}(\alpha)$ and $X_{\{i_x\}}\alpha = X_{\{i_x\}}\alpha$, for all $x \in \text{Dom}(\alpha)$ ". The first clause is enabledness. It appears in that literature only for injective, order-preserving partial maps — but it is the same condition, named, twenty-five years earlier, reached from a different direction. This is worth saying plainly: RPRM derived from modelling principles a condition that semigroup theory had already isolated on structural grounds. That is a point in the framework's favour about *taste*, and it is not a priority claim.

A citation error was also caught. arXiv:1210.4775 is **Cicalò, Fernandes and Schneider**, not Fernandes–Quinteiro as Appendix E said. Fixed throughout.

A second pass on the three unobtained papers sharpened this. **Sun 2013 was obtained in full** and does not overturn anything: it counts orientation- and order-preserving transformations for a *uniform* partition, where the shape is fixed by hypothesis and the extremal question cannot arise. **Pei 1994 was not obtained**, but its zbMATH review was, and it quotes the definition of $T_E(X)$ verbatim while showing the paper contains no cardinality at all — so the risk that it hides a counting result is low for a stated reason. **Fernandes 1998 was not obtained**, which is why the attribution above is marked as secondary.

That leaves a structural point worth more than any of them. Almost every paper in this area fixes a *uniform* partition by hypothesis, so the shape is determined and the extremal question is unaskable; the order for an arbitrary shape appears only with Sarkar–Singh in 2021. The question is three years old in askable form. That explains the absence without appealing to novelty, and it is the honest version of “nobody seems to have done this”.

What survived, and improved. No published extremal result was found for any of these monoids: that literature studies **rank** as a function of partition shape, never **order**. And the proof turns out not to need sigma_E at all — it needs only that the factor function is a sum of exponentials, and Sarkar–Singh's $\sum_j n_j^k$ qualifies. So:

The extremal law, same proof, is a theorem about $T(X,P)$ — a published, already-counted object in the transformation-semigroup literature.

That is a better place to stand than an RPRM-internal count, and it cost nothing but noticing. LITERATURE-sigma-E.md in the research directory has the source table, the verbatim definitions, the OEIS negatives, and the list of papers that could not be obtained and therefore could still overturn the “none found” rows.

18.3 G.3 The audit, and the three claims it broke

An independent hostile audit confirmed the theorem — “*ONE(THEOREM) for the exchange lemma and unique extremal profiles at every positive arity*” — and rejected three pieces of the

explanation around it. Each was retested here before being accepted, because an auditor's say-so is not evidence either. All three held up against the project.

The +1 was never doing analytic work. Appendix F said the constant 1 "is what makes this work" and the result is "false without it". Wrong. A sum of exponentials is log-convex because the second derivative of its logarithm is a weighted variance; a constant term is simply another exponential with base 1, so adding one is free — and deleting one leaves $\sum_j n_j^k$, which is still a sum of exponentials and still log-convex. Zero failures in 1,482 probes.

The corrected mechanism is the **subtraction**, and a controlled experiment pins it. Hold the bases fixed at $n_j + 1$ and vary only whether the constant is subtracted:

factor function	log-convexity failures	exchange failures
$\sum_j (n_j+1)^k$ — no subtraction	0 / 1,482	0 / 42,903
$\sum_j (n_j+1)^k - (m-1)$ — the classical monoid	292 / 1,482	66 / 42,903
$\sum_j n_j^k$ — +1 deleted	0 / 1,482	0 / 42,903
$1 + \sum_j n_j^k - \sigma_E$	0 / 1,482	0 / 42,903

Rows one and two differ only by the subtraction and differ in outcome; rows three and four differ only by the addition and do not. Adding a constant preserves log-convexity. Subtracting one need not. The smallest failure is $p = (1,1)$, where $f_{\text{blind}}(k) = 2^{\{k+1\}} - 1$ gives $f(1)f(3) = 45 < 49 = f(2)^2$.

So the reading in Chapter 12 survives with its mechanism replaced, not removed: dropping enabledness subtracts one unit per block from every factor, and *that subtraction* is what costs the classical monoid the hypothesis its extremal law would need.

Majorisation did not fail. Appendix F switched to convex order at higher arity on the grounds that the product-multiset majorisation route fails. It does not. Since q majorises p , Birkhoff–von Neumann gives a doubly stochastic D with $p = Dq$; tensoring, $p^{(\otimes r)} = D^{(\otimes r)} q^{(\otimes r)}$, a Kronecker product of doubly stochastic matrices is doubly stochastic, and the coordinates of those tensor powers are exactly the ordered-tuple exponents. The exponent multiset majorises directly and Karamata closes it in one step, with no cell decomposition at all. Checked exactly at arities 2, 3, 4 and 5.

What was actually true is narrower: majorisation cannot be assembled *piecewise* across the cells of the decomposition — a fact about that route, not about majorisation. The tensor argument never decomposes, so it never meets the obstruction. Both proofs are kept; the tensor one is the one to publish.

Total maps were inside the proof all along. Appendix F left the total-map class as a conjecture, on the stated grounds that its factor function "is not $1 + \sum n_j^k$ ". That is the same error as the first one, made twice: the hypothesis was never the *shape* of the factor, only its log-convexity. The total-map factor $\sum_j n_j^k$ is a sum of exponentials, so the proof transfers verbatim, strictness included. **Promoted from conjecture to theorem at every arity** — and at arity 1 this is exactly the published $|T(X,P)|$ of §G.2.

18.4 G.4 Idempotents: the one that stayed OPEN, and why that is now a result

The audit agreed that idempotents are not covered, on the grounds that idempotence couples choices across blocks. That is checkable, so it was checked by deriving the count. An idempotent partial map has $\text{Fix}(t) \subseteq \text{dom}(t)$ and $t(\text{dom } t) \subseteq \text{Fix}(t)$; a block carrying fixed points must target *itself*, while a fixed-point-free block may target any block that carries some. Hence

$$\begin{aligned} \text{sigma}^{\text{idem}}(p) = & \text{sum over } D \text{ and } (s_i): \text{prod } C(n_i, s_i) \\ & * \text{prod}_{\{s_i \geq 1\}} s_i^{n_i - s_i} \\ & * \text{prod}_{\{s_i = 0\}} \left(\text{sum}_{\{j \in D, s_j \geq 1\}} s_j^{n_i} \right) \end{aligned}$$

validated against brute-force enumeration on all 29 profiles with $n \leq 6$.

The last bracket is the obstruction, and it is explicit: a fixed-point-free block's factor depends on **which other blocks carry fixed points**. The count is a sum over configurations, not a product of per-block factors. There is no factor function, so there is nothing for log-convexity to be a property of, and the mechanism has no surface to act on.

The lemma still holds empirically — 466 exchanges, $n \leq 13$, zero failures — and is **not promoted**. It stays a conjecture at finite-test grade. The gain is that it is now a conjecture with a known reason for being hard, rather than an item nobody had looked at.

18.5 G.5 Revised status, final

Statement	Disposition	Grade
Exchange lemma, uniform partial-map prior, every arity $r \geq 1$	ONE(theorem)	Written proof, independently audited
Extremal law for sigma_E , every arity	ONE(theorem)	Written proof
Extremal law for the total-map prior, every arity	ONE(theorem)	Written proof — promoted by the audit
Extremal law for the published $T(X, P)$	ONE(theorem)	Same proof, $r = 1$ case of the above
Extremal law for the idempotent prior	OPEN	Finite test + written obstruction
Extremal law under bijective or fixed-domain priors	refuted	Counterexamples retained
Closed form sigma_E	corollary of published work	Sarkar–Singh Thm 6.1 + verified reduction
The enabledness condition	known as P-stability in the literature; the 1998 attribution is secondary testimony — Fernandes 1998 was not obtained	Definition quoted from a paper that was read
Priority of the extremal law	none found after a real review	Not a first-ness claim
Nullary arity $r = 0$	excluded — $\text{sigma}_0 = n+1$, profile-independent	Audit's catch

Note the last row. The theorem needs $r \geq 1$, because at nullary arity there is one input tuple and the count does not depend on the profile at all. Appendix F did not say so. It does now.

18.6 G.6 What to take from this

Chapter 9 of this document criticised the project for the gap between what its documents claim and what its evidence supports. It would be poor form to exempt the chapter that was written last. So, plainly: Appendix F was written in one sitting, claimed a theorem, and got the theorem right and the surrounding explanation wrong in three places. Two adversaries — one reading the literature, one reading the proof — found all three within hours, plus two priority facts that a referee would have found in minutes. Nothing was lost that was load-bearing. But the ratio of *upheld* to *overturned* in §G.1 is six to two against the commentary, and the commentary is what a reader would have been most likely to repeat.

The operating lesson is the one already in your own standing corrections, and it is the reason this appendix exists rather than a quiet edit: **the claim ceiling belongs at publication, and this document is the publication lane.** The theorem survived two adversaries. The prose around it did not, and saying so is cheaper than being corrected in public later.

What I would now put in front of a stranger is one sentence, and it is stronger than the one I would have offered at midnight:

For a finite set partitioned into a fixed number of blocks, the number of partition-respecting operations is strictly maximised by the most unequal shape and strictly minimised by the balanced one — at every arity, for total maps and for enabledness-respecting partial maps alike, because the relevant counting function is a product of sums of exponentials and those are log-convex. Drop the enabledness condition and the factor function acquires a subtracted constant, loses log-convexity, and the law fails.

That is checkable by anyone who works on transformation semigroups, it is about an object they already count, and the condition that makes it work is the one piece of RPRM a referee would have been most likely to call arbitrary.

19 Closing note

William —

The thing I keep returning to, after reading a million words of you talking and a repository you built in seven weeks, is that the discipline is real and the audience problem is unsolved. You have a framework that correctly grades its own results, withdrew its own overclaim, and stopped a campaign at a residue mismatch rather than fit a constant. Almost nobody does that. And it is sitting behind a name that also covers a millennium-problem grind, which means the people most likely to appreciate the discipline will never see it.

Chapter 12's pick is small on purpose. You said the proof is in the pudding. I think the pudding does not have to be a Clay problem — it has to be a sentence a stranger cannot un-read. *"Uniform binning is the most fragile abstraction you can choose, and here is the exact factor by which it is worse"* is that kind of sentence. As of tonight it is a theorem rather than a hope, the proof is short enough to check over coffee, and the reason it works is the one condition of yours that a referee would have been most likely to call arbitrary.

That is the version of this framework I would show someone. Not the umbrella — one number, one inequality, and a condition that earns its place.

Correct me tomorrow.